

063-0605-22L

Computational Structural Design I

Computational Graphic Statics

Dr. Lluís Enrique

Block Research Group (Prof. Dr. Philippe Block)

HS2022

Week 4 Friday, October 14th

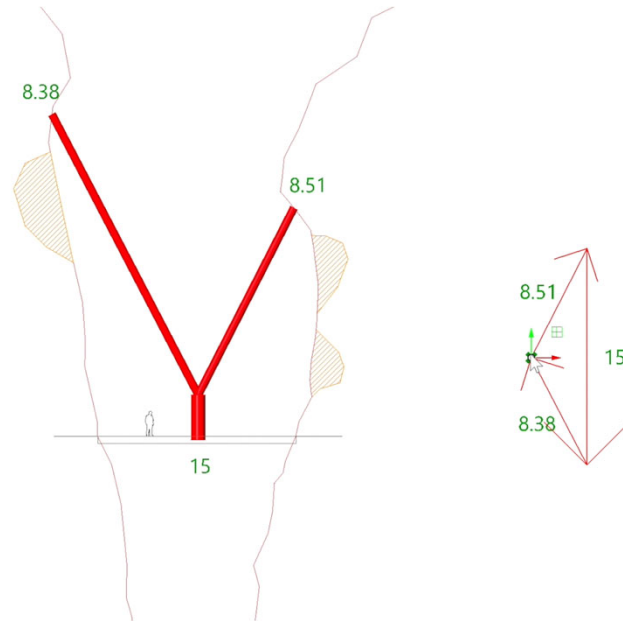
9:45 – 10:30 Recap of last week's exercise
Lecture: Computational Graphic Statics

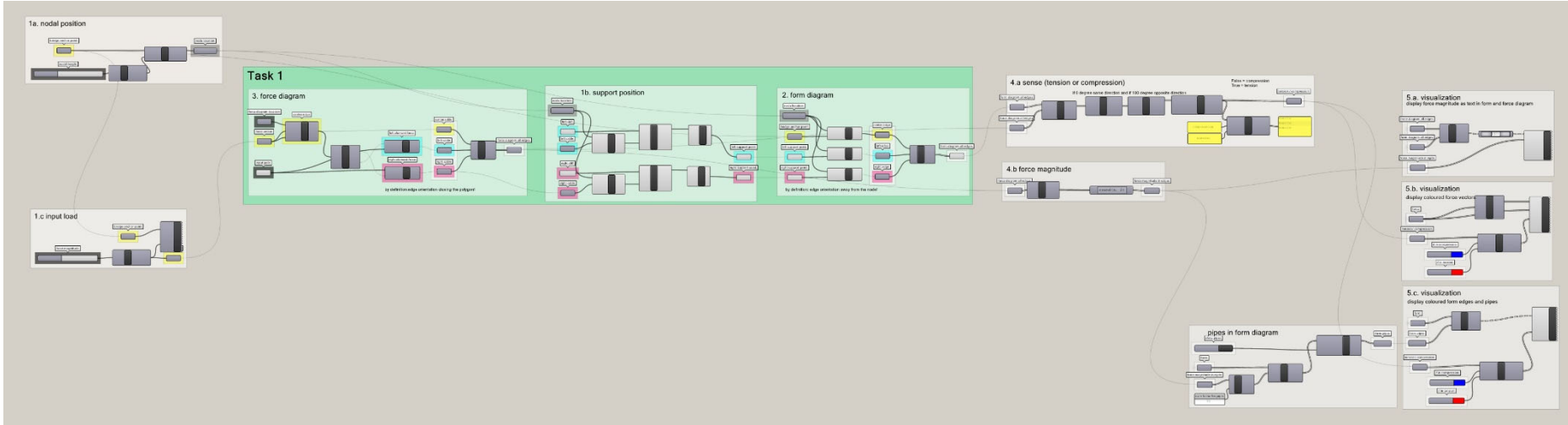
10:30 – 10:45 Break

10:45 – 11:30 **Tutorial: Procedural graphic statics II**

11:30 – 11:45 Break

11:45 – 12:30 **Tutorial: Procedural graphic statics II**

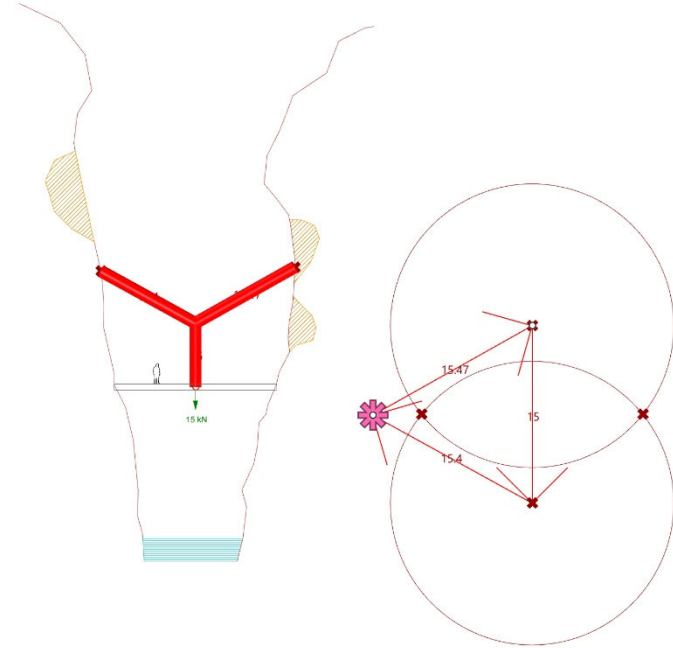
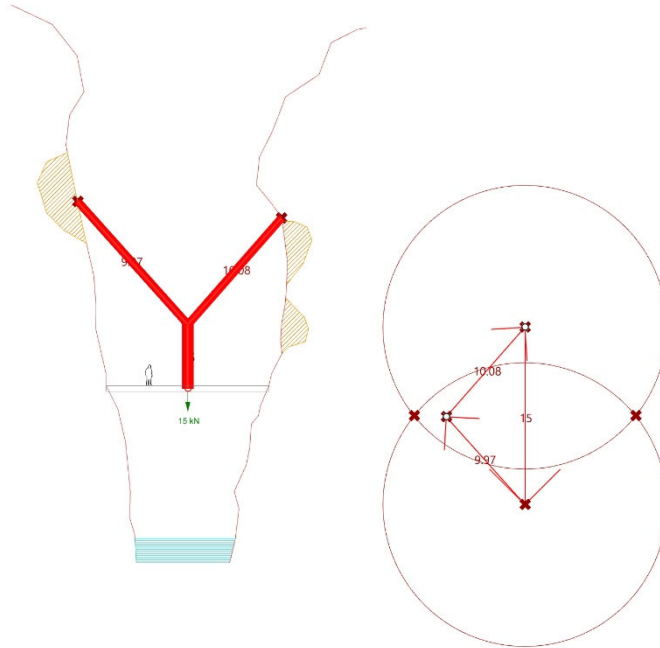


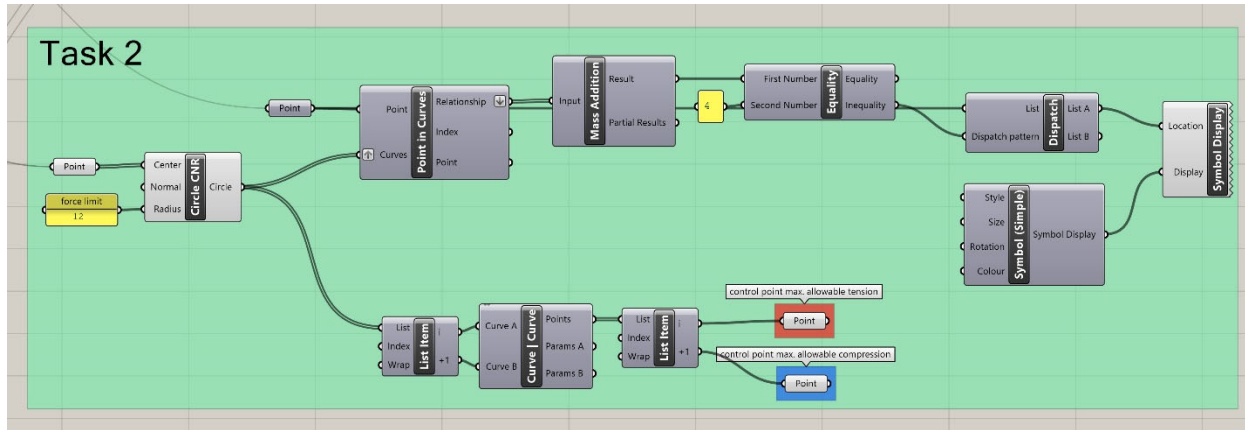


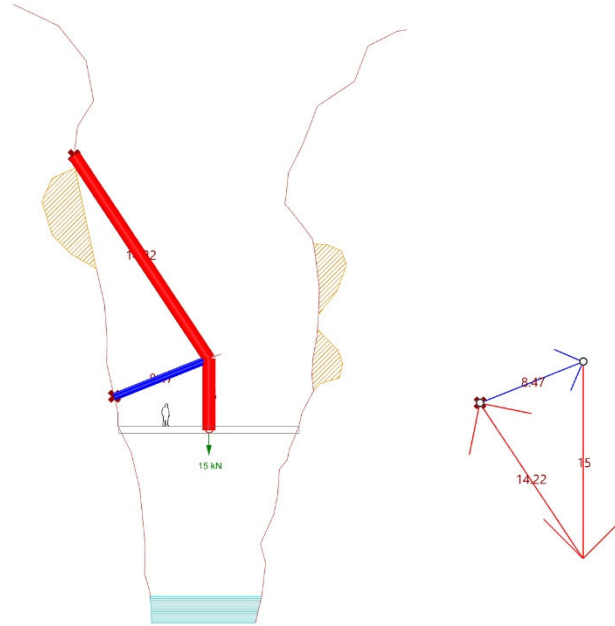
3. force diagram

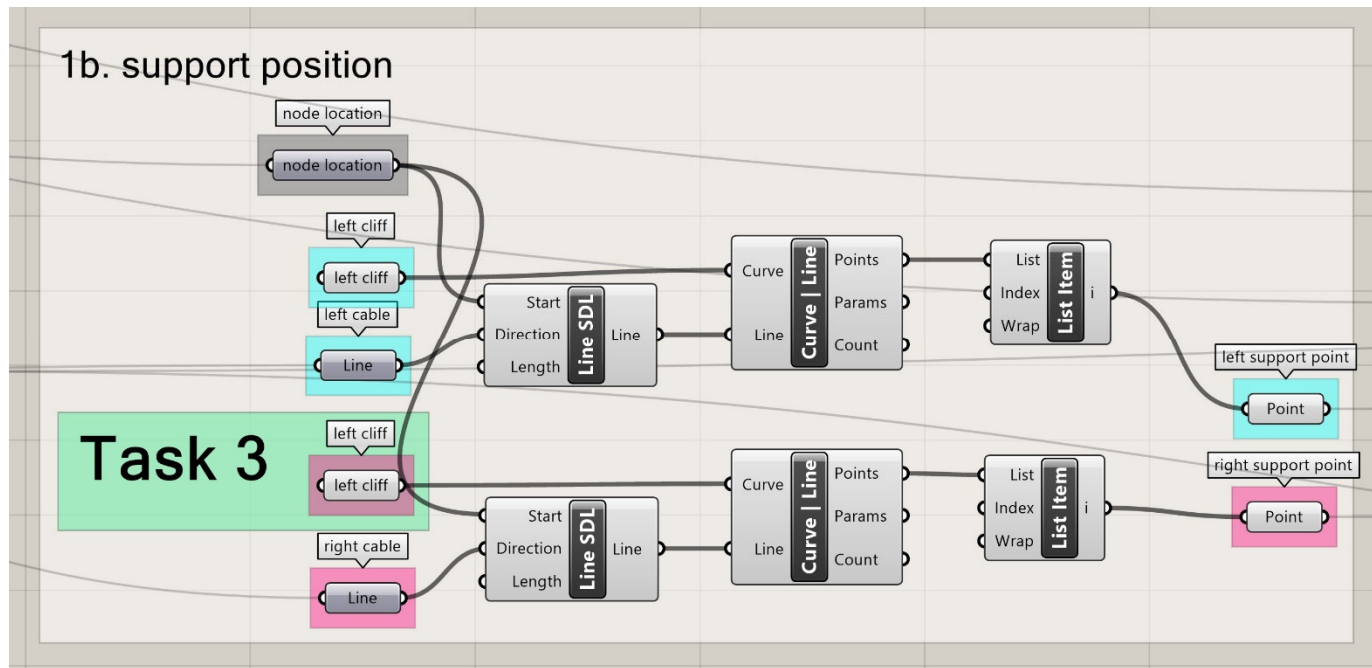


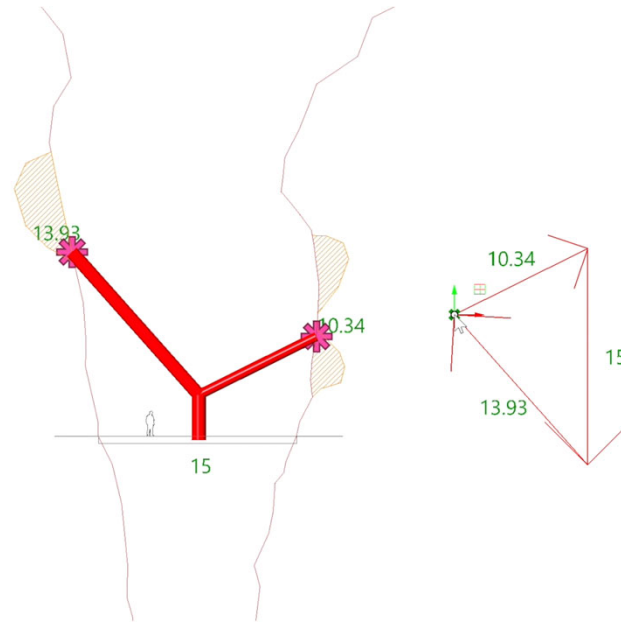
by definition: edge orientation away from the node!

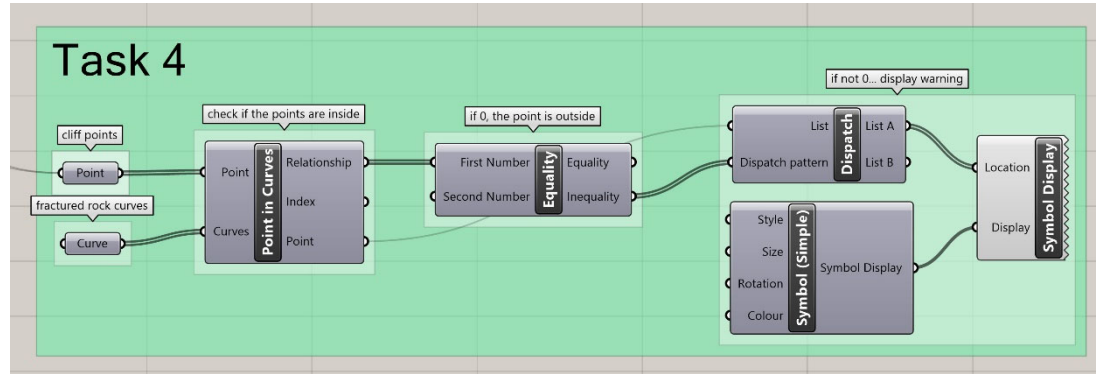


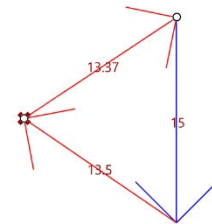
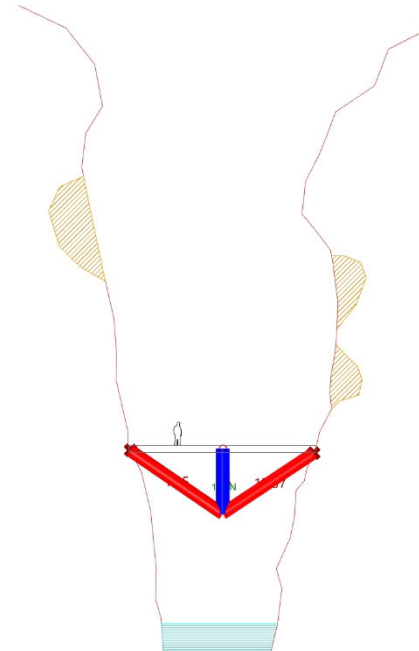
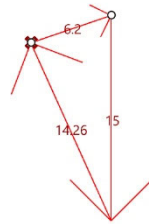
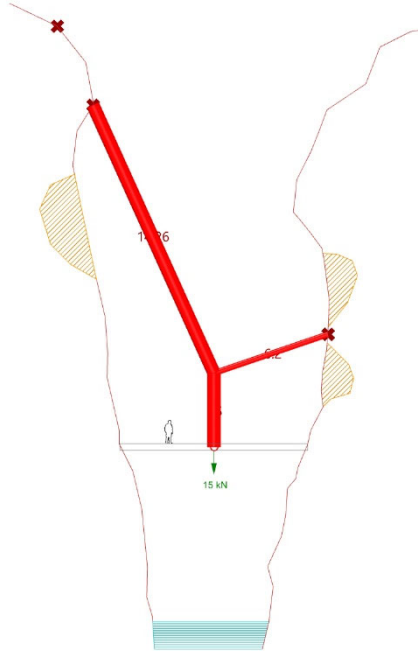




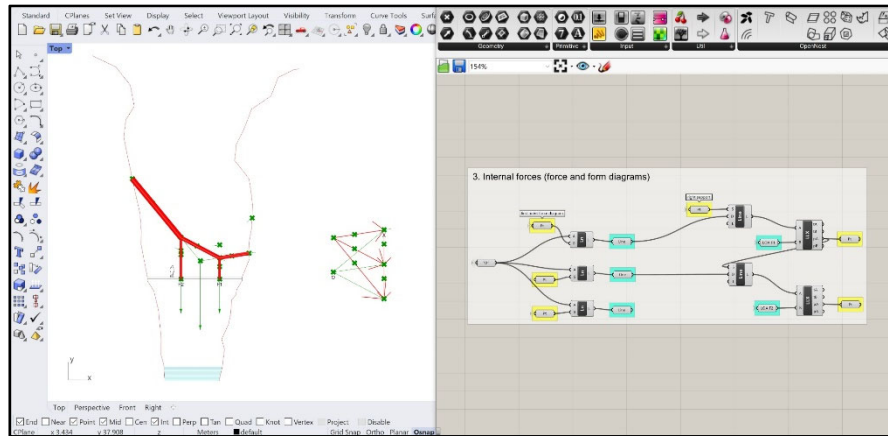
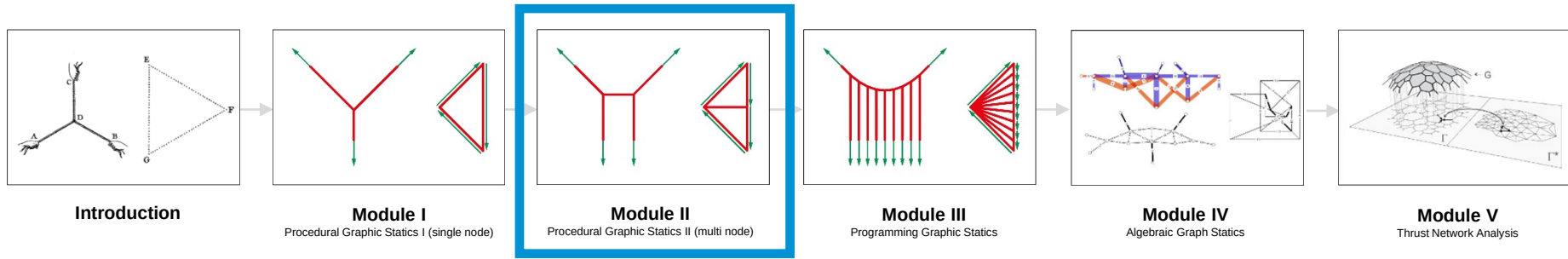








Typology	Topic	Module	Week		Topic	Lead				
cables & arches	Introduction		Week 1 (23/9)	Lecture	Introduction to graphic statics	Dr. Lluís Enrique				
				Tutorial	Introduction to Rhino and Grasshopper					
				Work session	graphic statics + Grasshopper					
	Procedural GS (Grasshopper)	I	Week 2 (30/9)	Lecture	Algorithmic design & thinking	Dr. Lluís Enrique				
				Tutorial	Single node bridge in Grasshopper					
			Week 3 (7/10)	Work session	Single node bridge in Grasshopper					
				II	Week 4 (14/10)		Lecture	Computational graphic statics	Dr. Lluís Enrique	
		Tutorial	Multi-node bridge in Grasshopper							
		Week 5 (21/10)	Work session		Multi-node bridge in Grasshopper					
			Week 6: Seminar week							
			cables & arches		Procedural GS (Python)	III	Week 7 (4/11)	Lecture		Programming
Tutorial		Multi-node bridge in Python								
Week 8 (11/11)	Work session	Multi-node bridge in Python								
trusses	AGS	IV	Week 9 (18/11)	Lecture	Algebraic Graph Statics (AGS)	Dr. Lluís Enrique				
				Tutorial	Interactive Graphic Statics (IGS)	Chaoyu Du				
			Week 10 (25/11)	Work session	Truss analysis using IGS					
				shells	TNA	V	Week 11 (02/12)	Lecture	Thrust Network Analysis (TNA)	Dr. Lluís Enrique
Tutorial	RhinoVault 2 (rV2)	Selina Bitting								
Week 12 (09/12)	Work session	Shell design using rV2								



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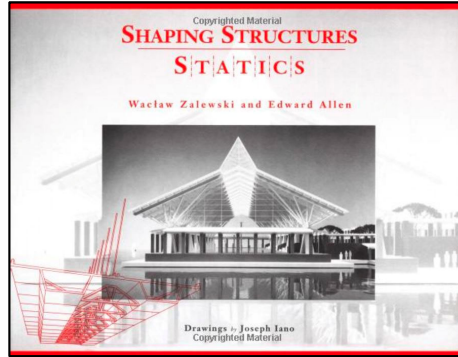


Computational Graphic Statics

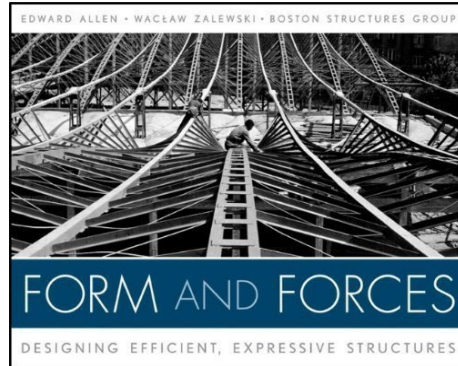
Friday, October 14th, 2022

Dr. Lluís Enrique

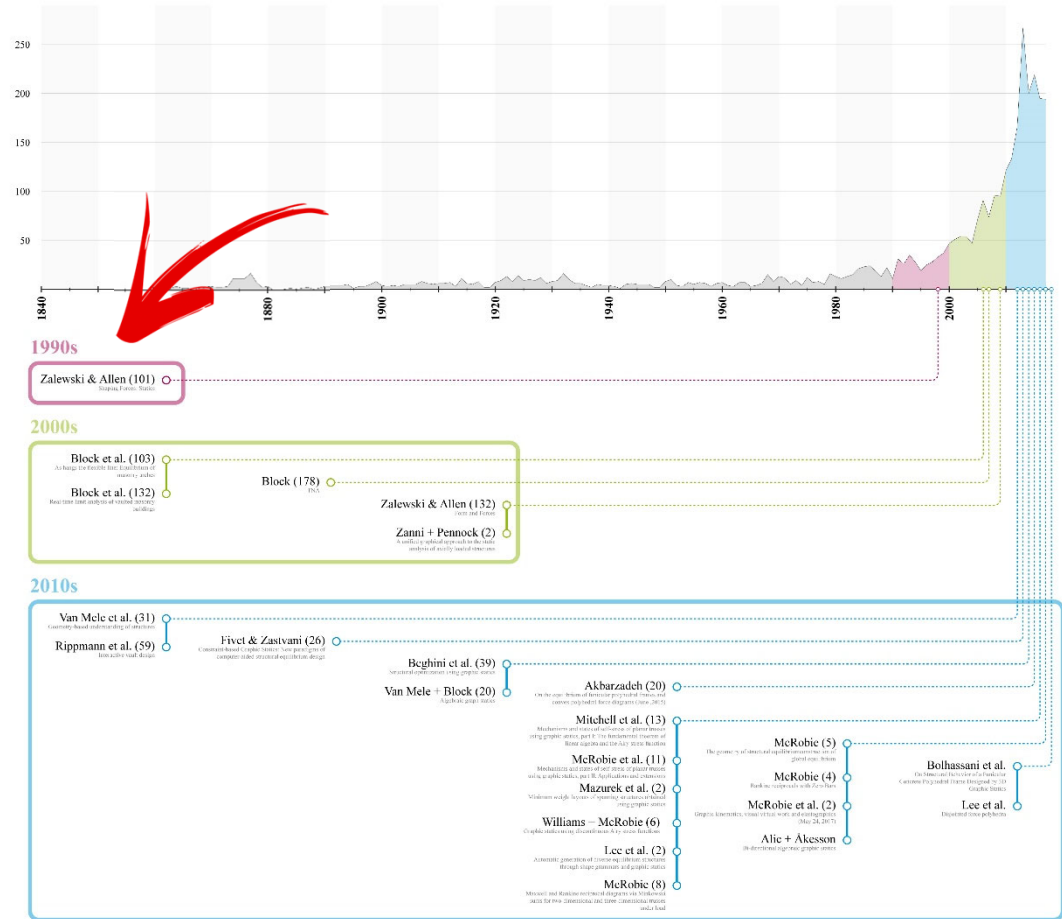
Graphic Statics

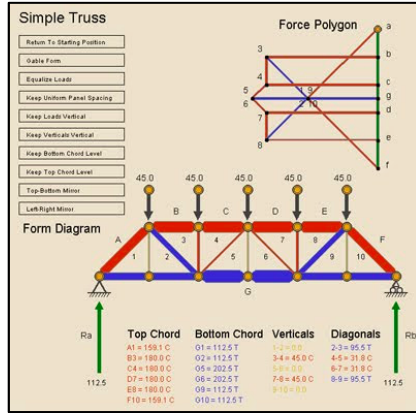


Shaping Structures
Zalewski and Allen (1997)



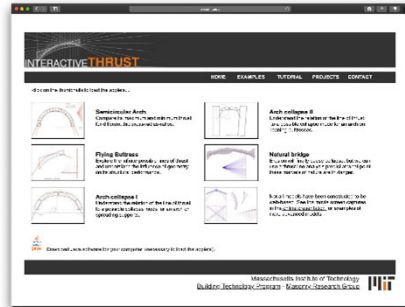
Form and Forces
(Allen and Zalewski (2010))





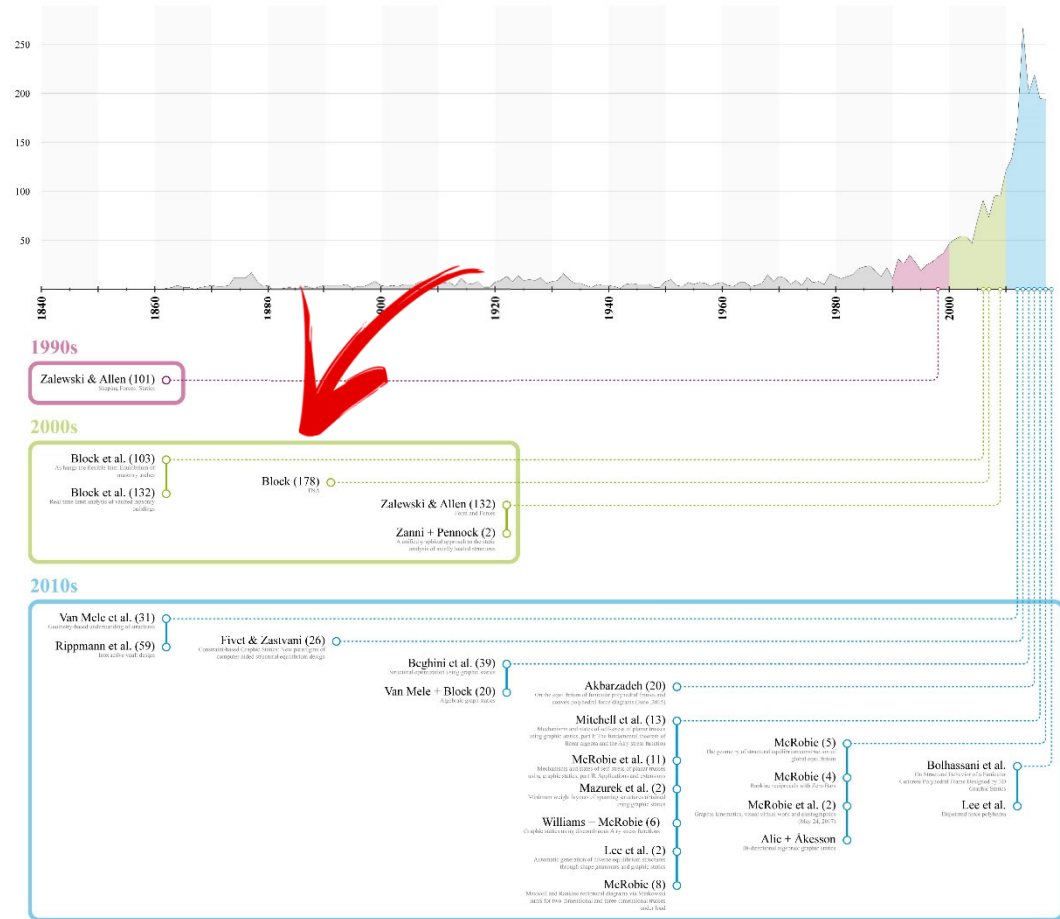
Active Statics

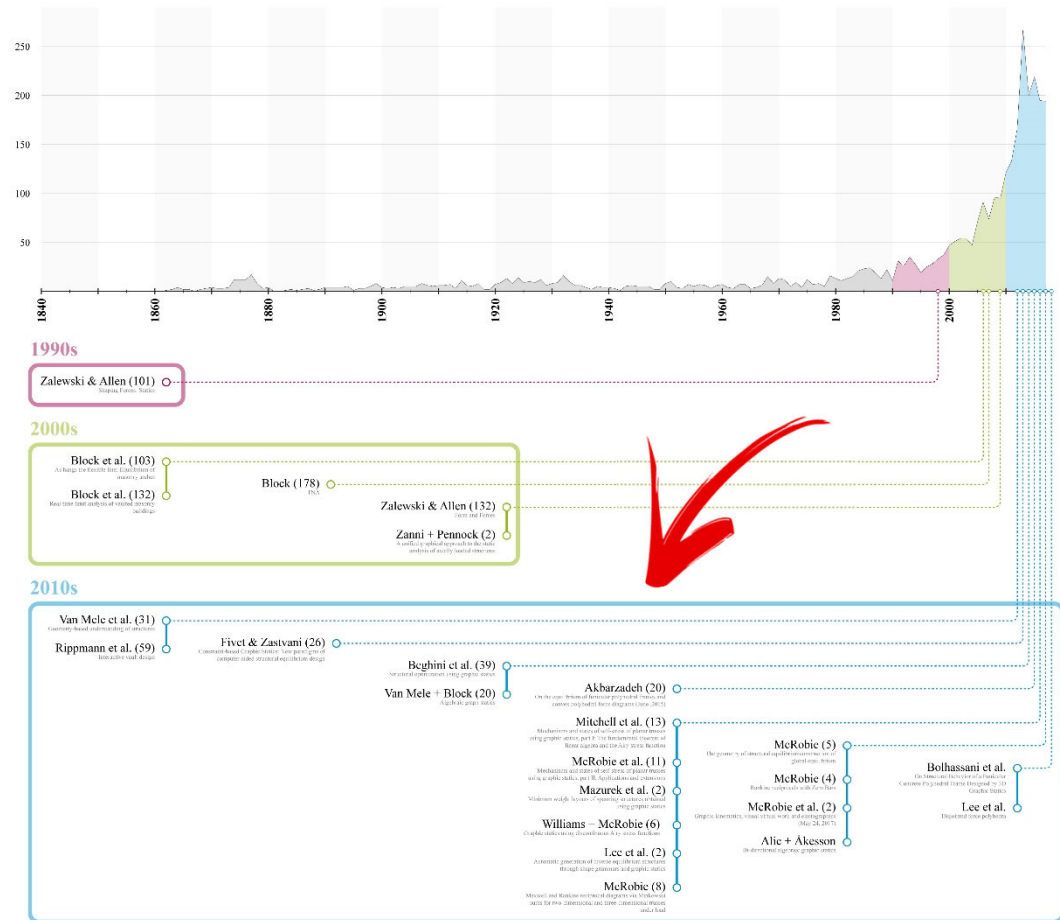
Greenwold and Allen (2003)

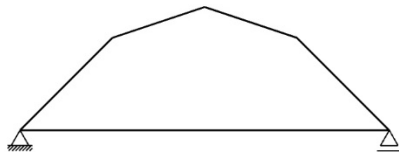
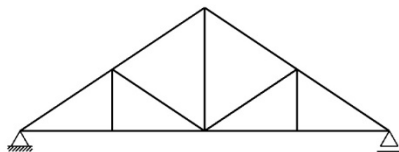
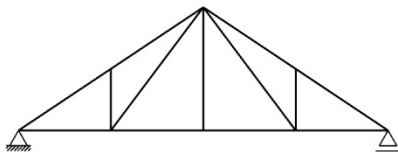
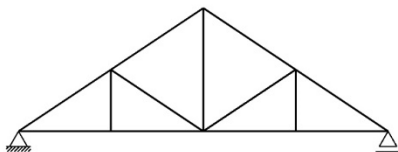
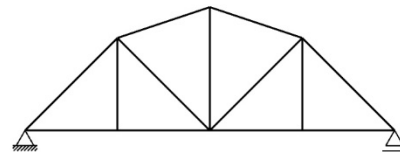
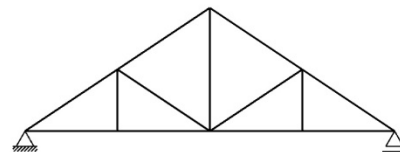


InteractiveTHRUST

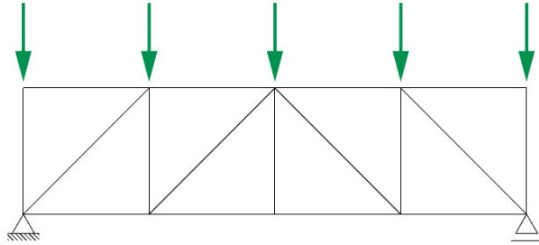
Block and Ochsendorf (2005)





Typology**Topology****Geometry**

Analysis



Input:

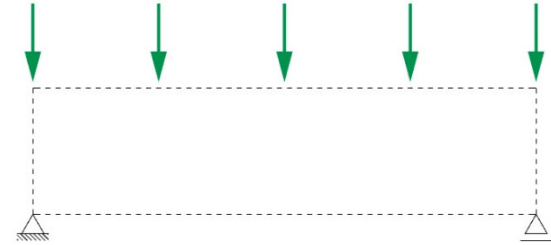
- Loads
- Supports
- Geometry



Trusses
3D trusses



Form-finding

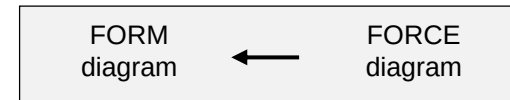


Input:

- Loads
- Supports



Cables
Arches
Arch-cables
Cable nets
Shells



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)

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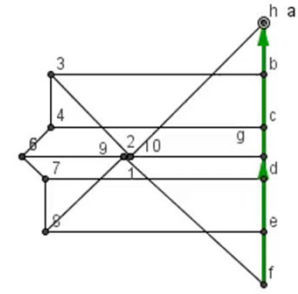
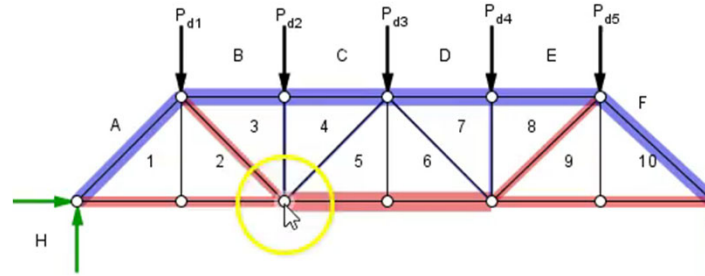
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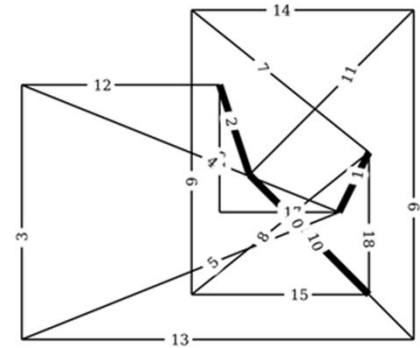
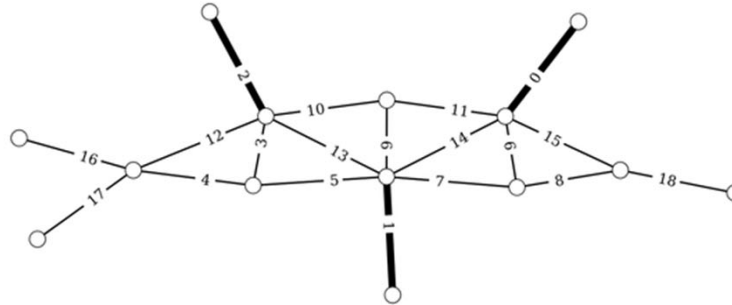
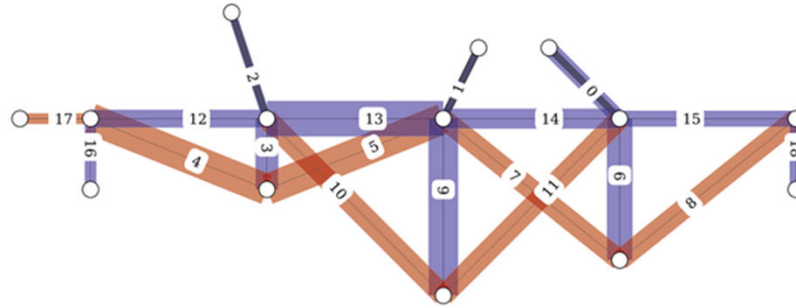
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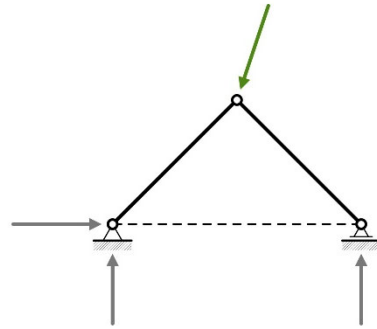
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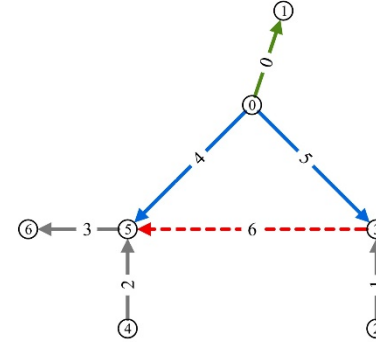
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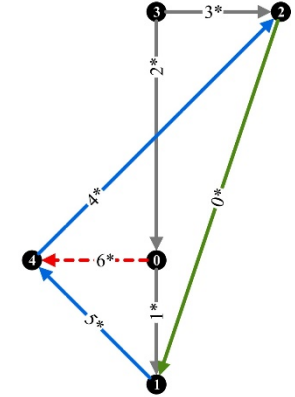
Combinatorial Equilibrium Modelling (CEM)



Form diagram



Form diagram as a graph



Force diagram

Get topology and connectivity

Solve via equilibrium matrix

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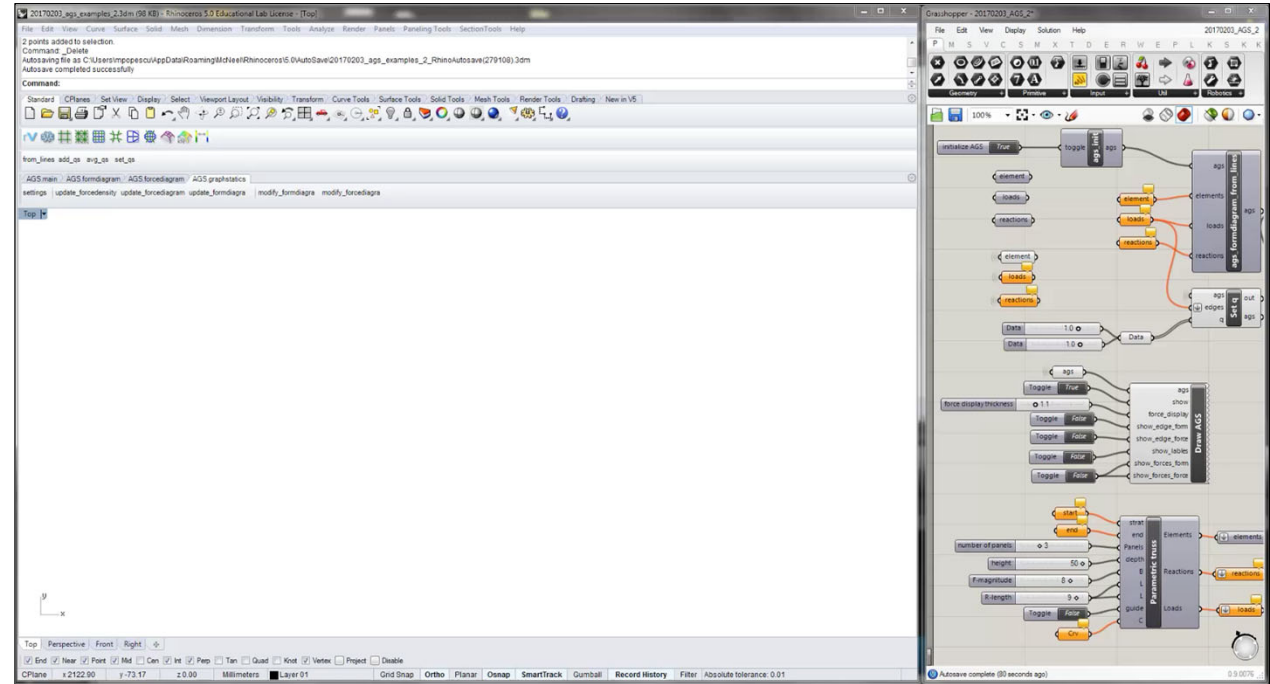
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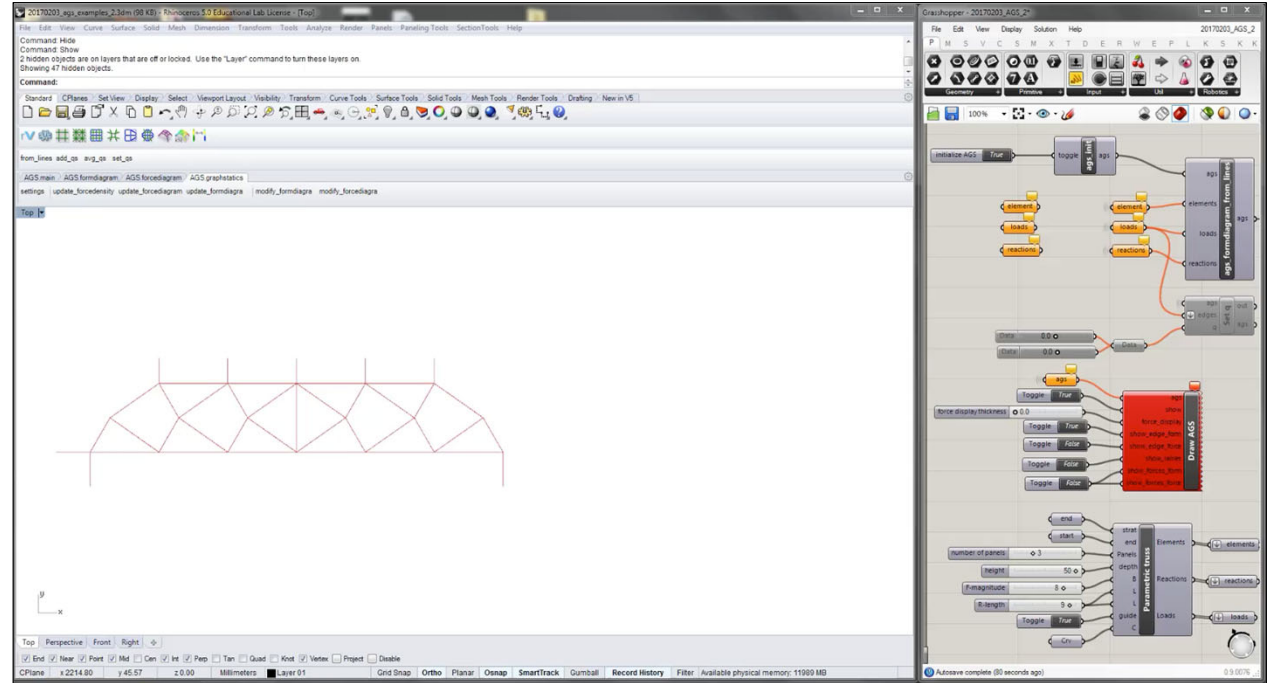
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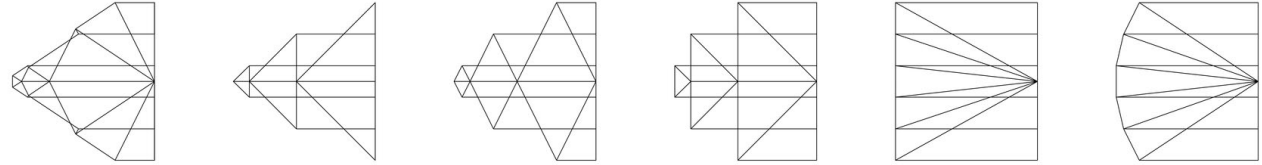
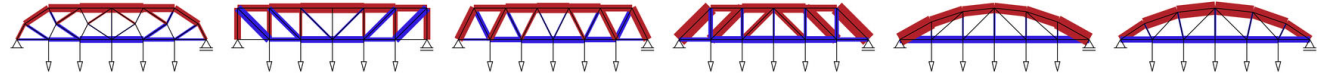
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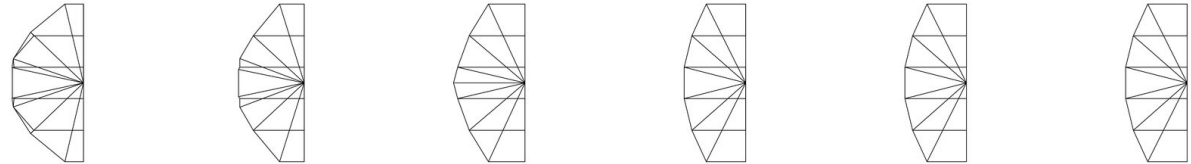
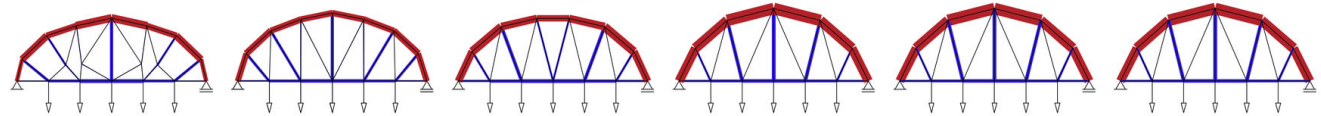
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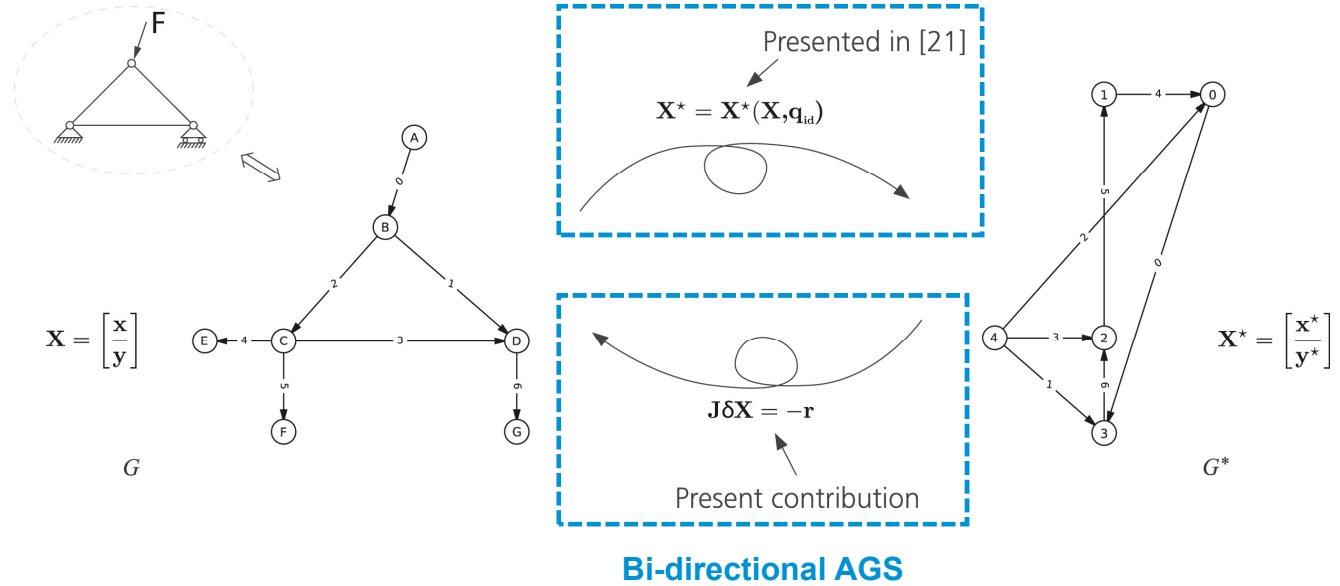
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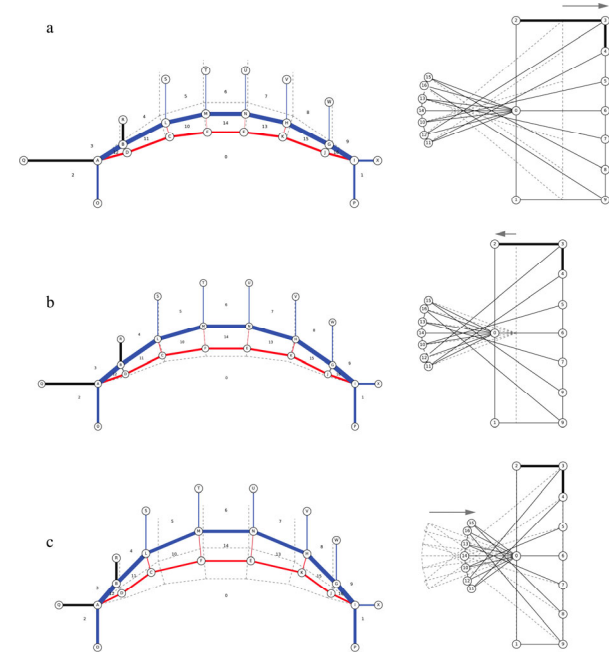
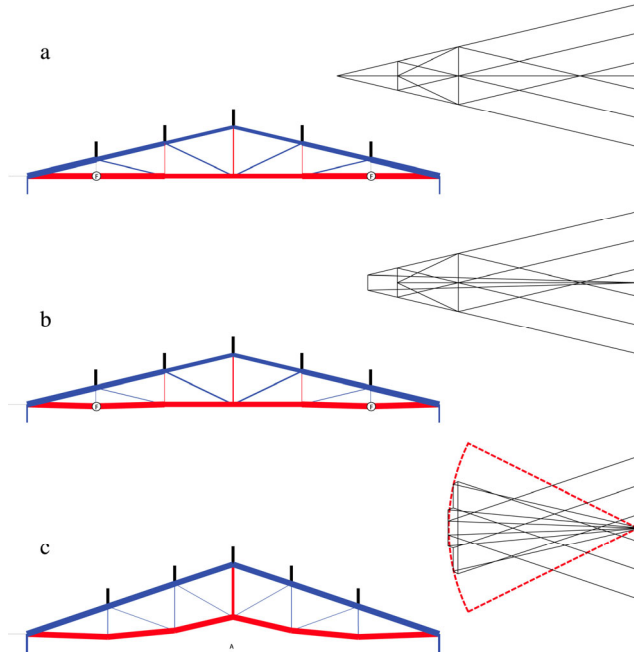
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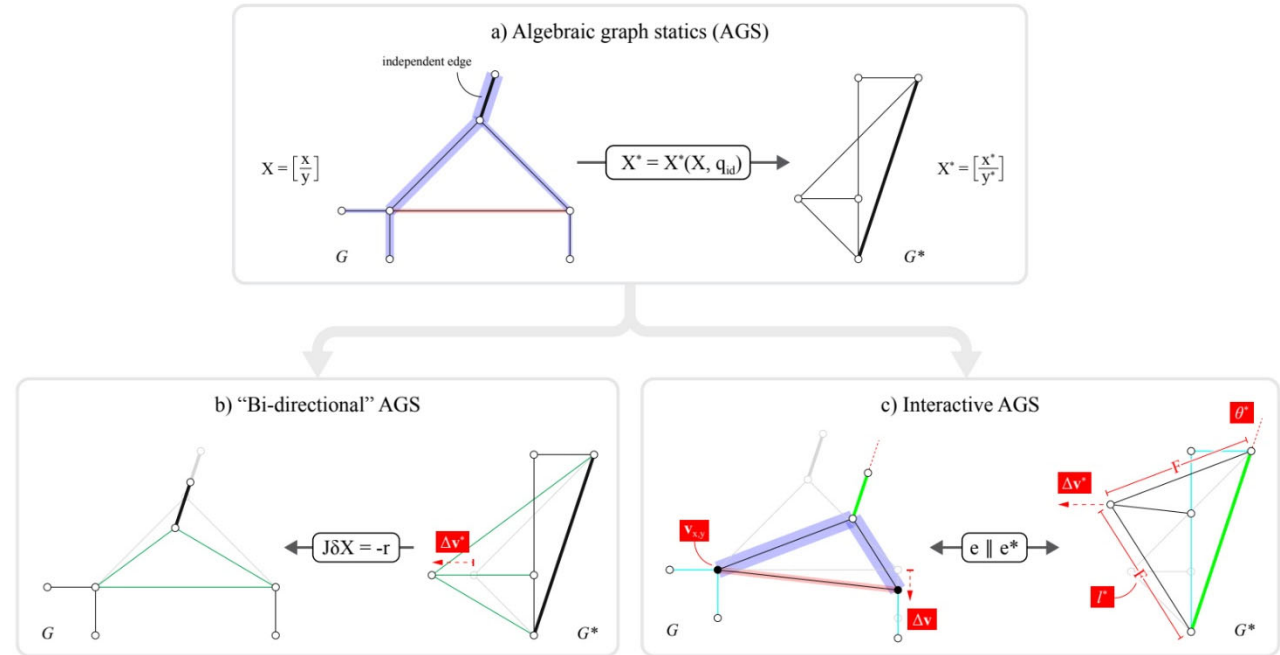
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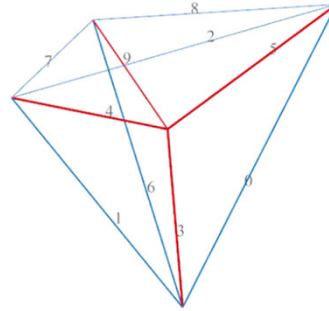
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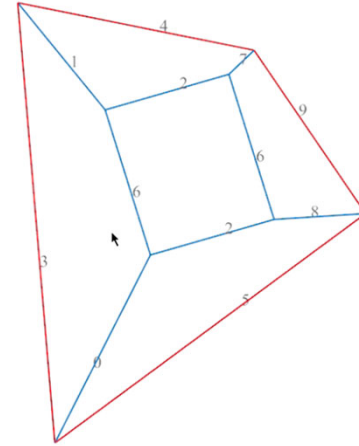


F



3D form diagram

F**'



vector-based
3D force diagram

— tension
— compression

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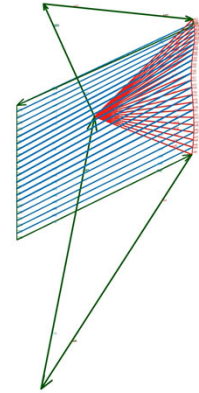
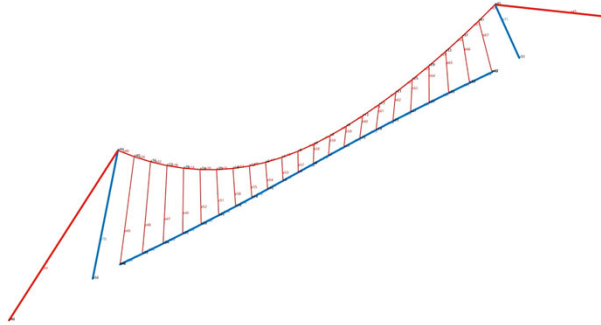
Shape grammars

Combinatorial Equilibrium Modelling (CEM)



F

F*



— tension
— compression

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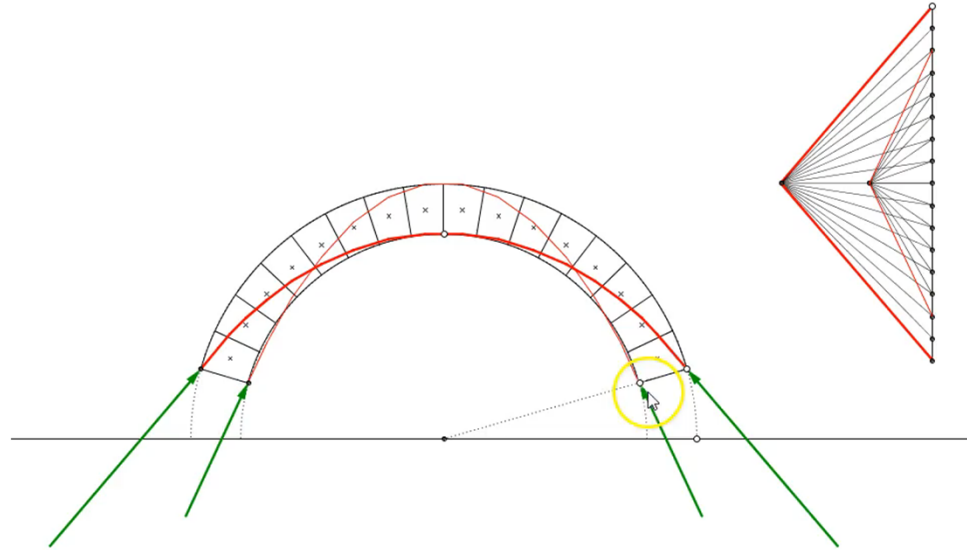
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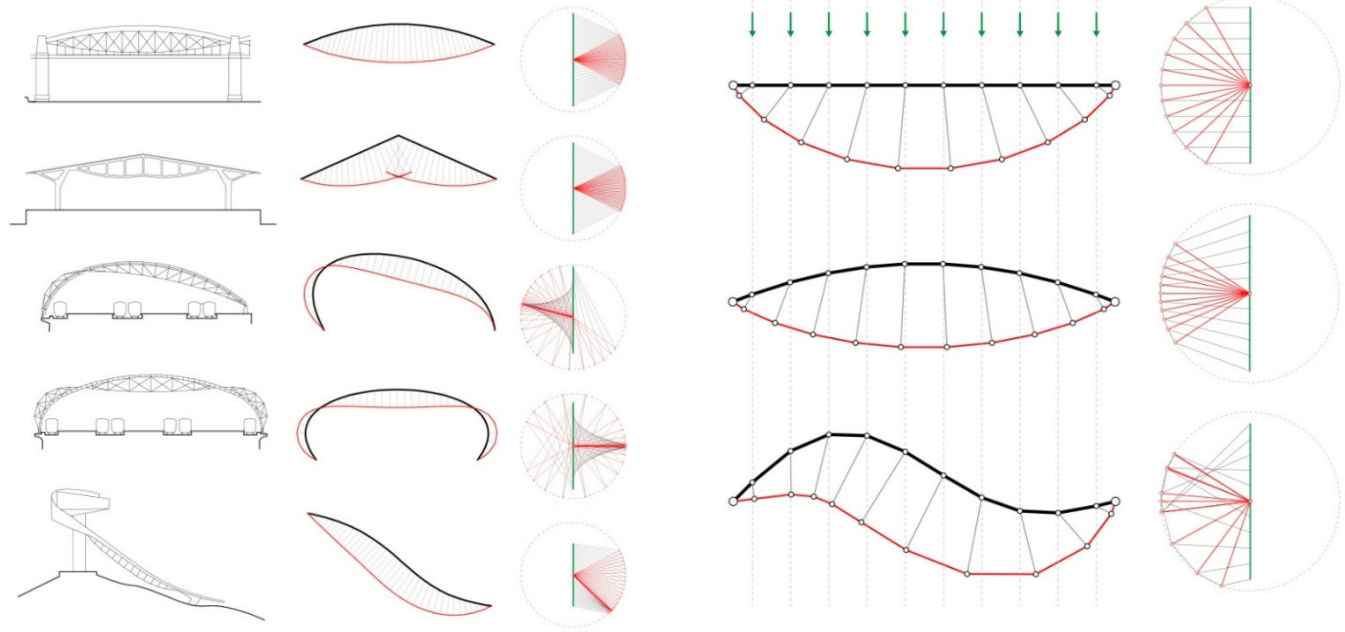
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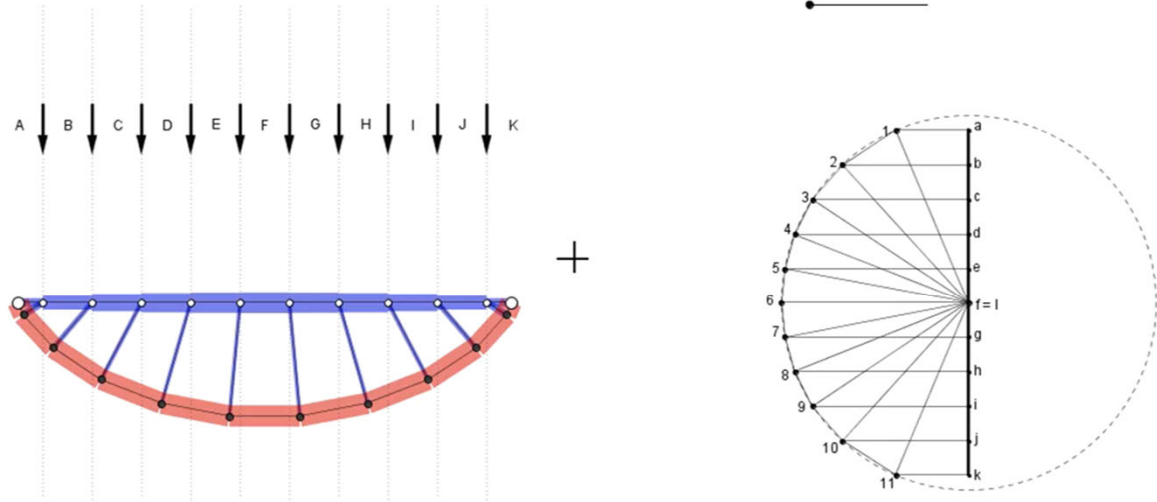
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[Constrained form-finding](#)

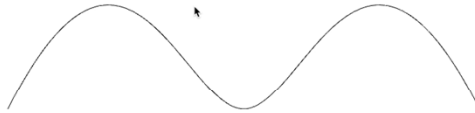
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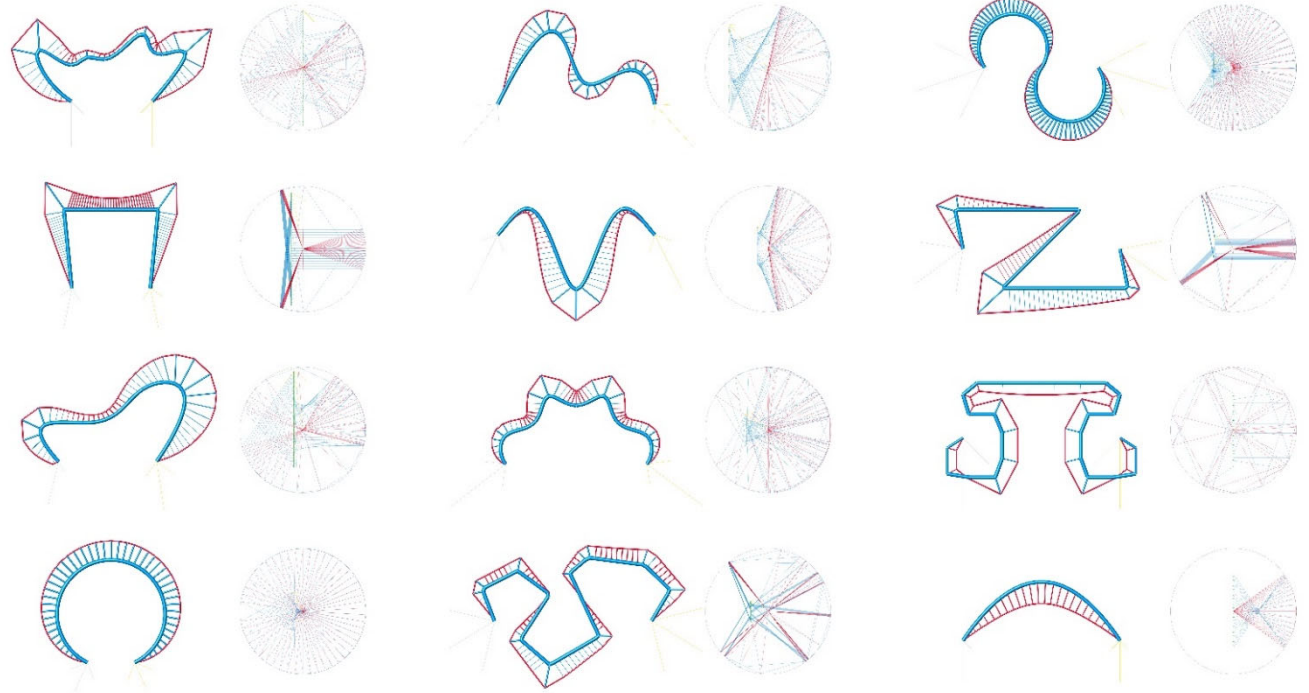
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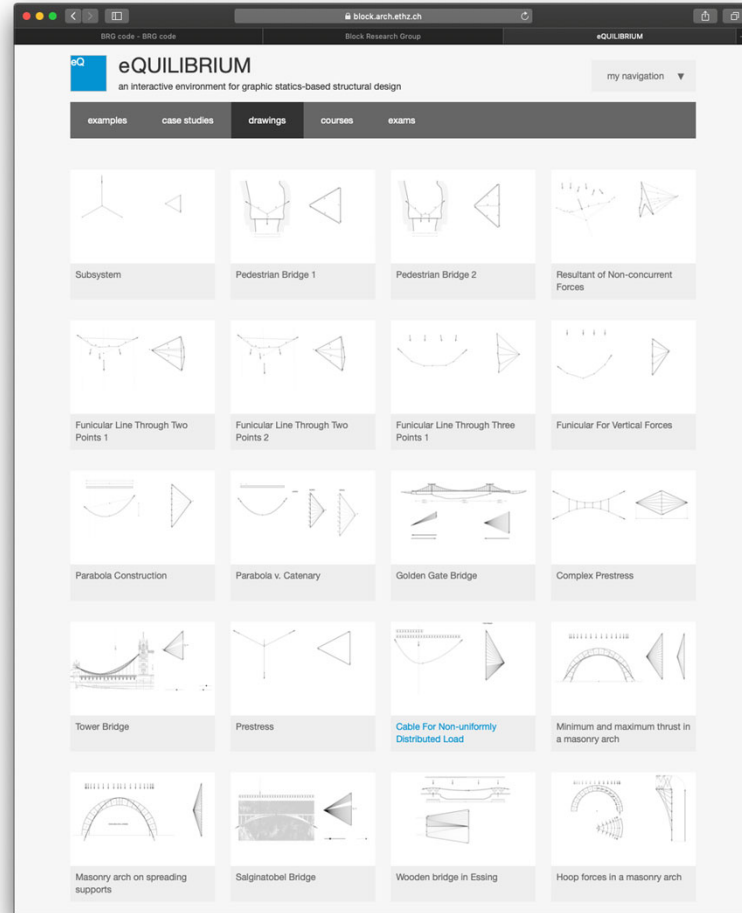
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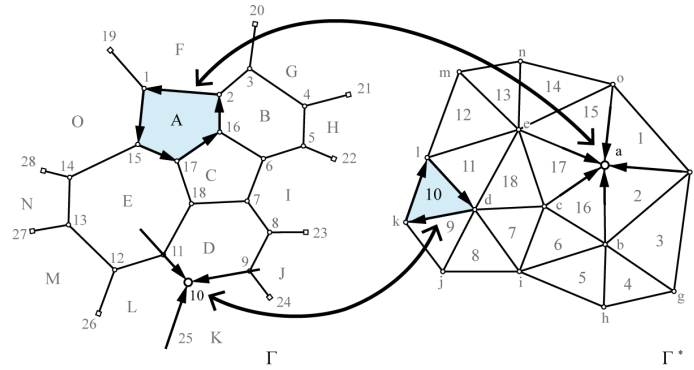
[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

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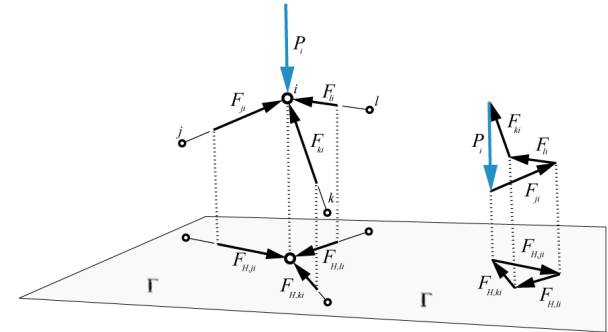
Shape grammars

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Form Diagram

Force Diagram



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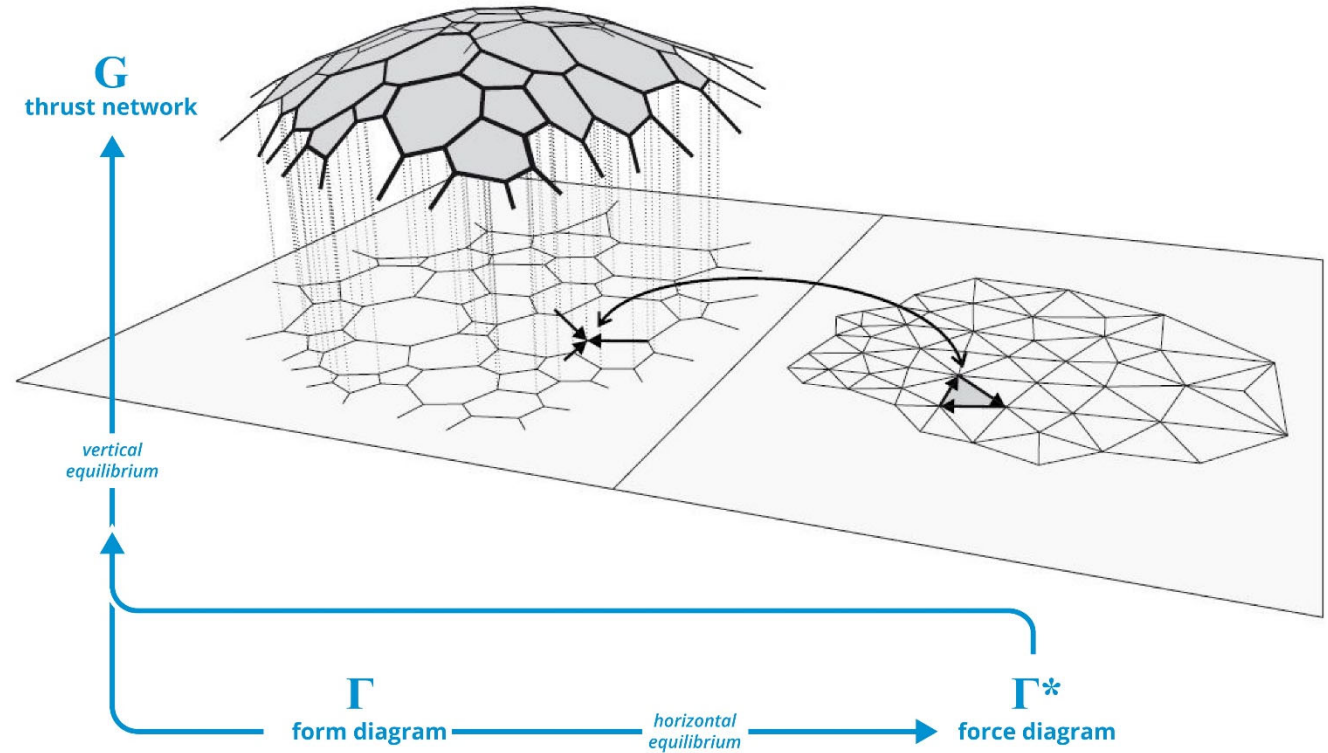
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Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

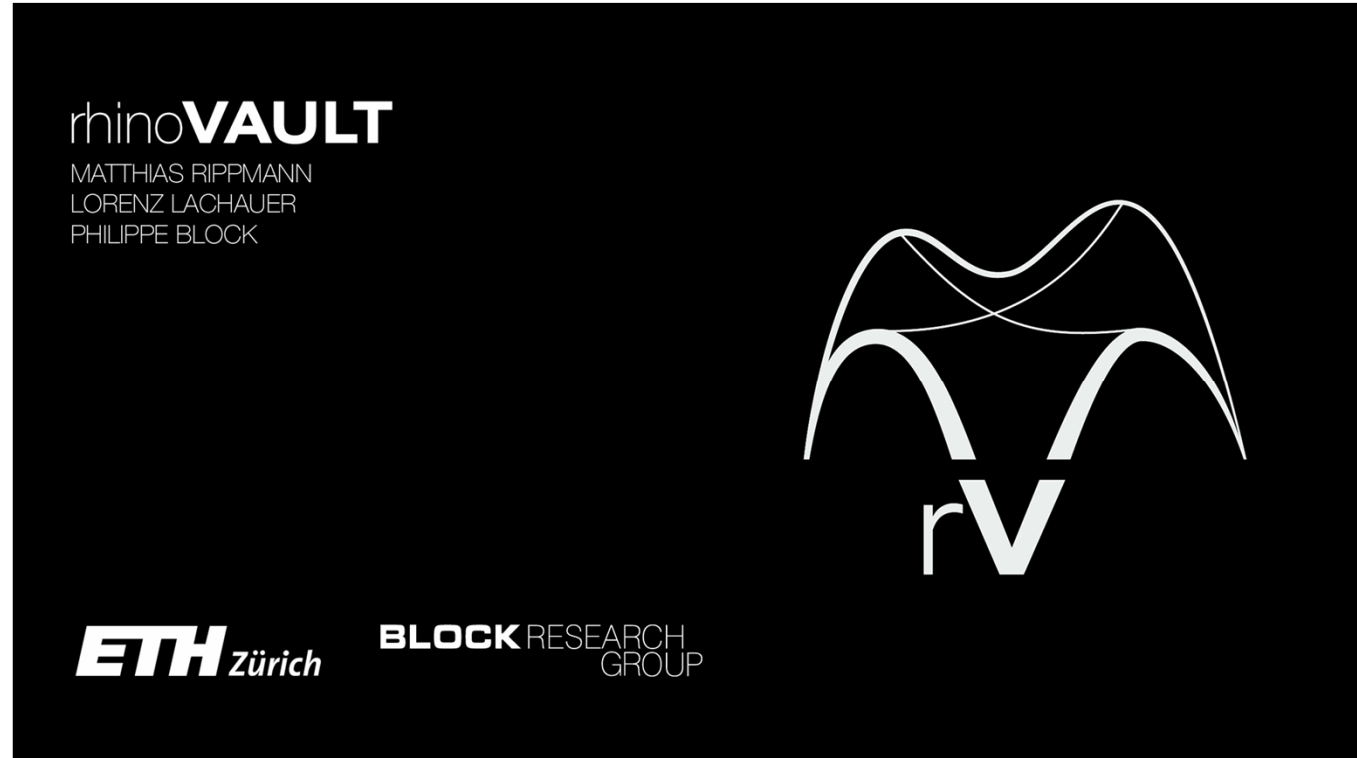
[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

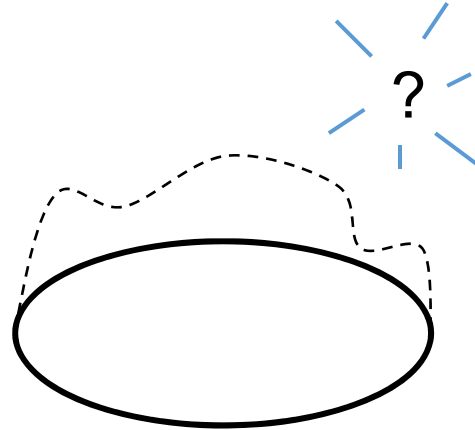
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3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Form-finding

Interactive drawings

Constrained form-finding

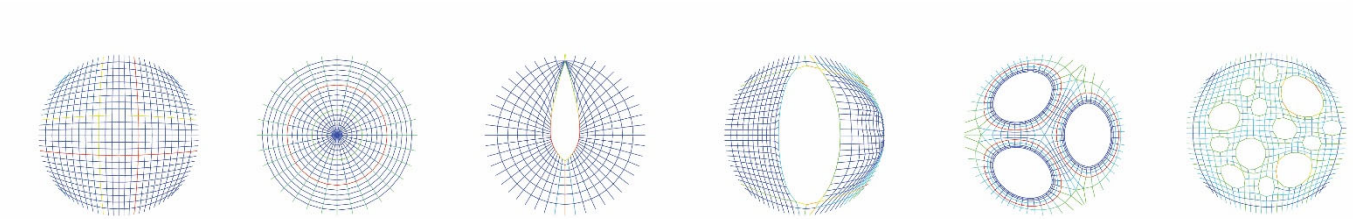
[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



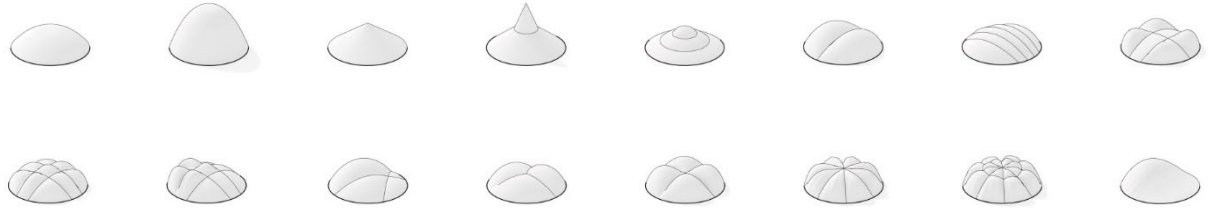
Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)



Form-finding

Interactive drawings

Constrained form-finding

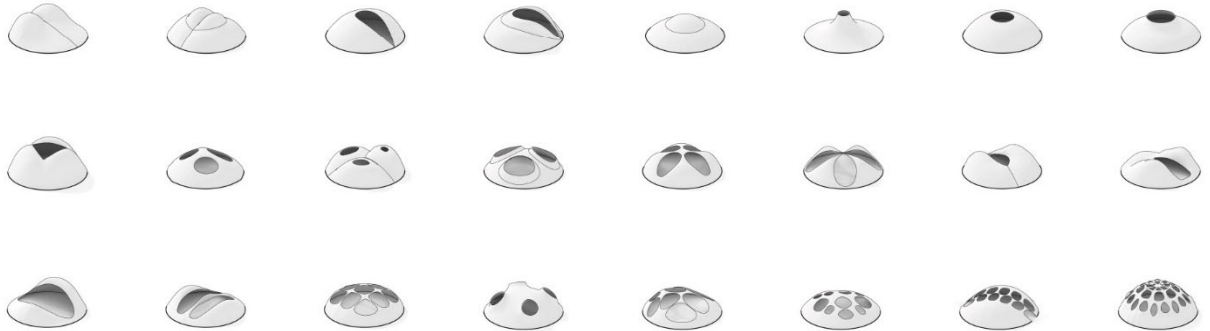
[Thrust Network Analysis \(TNA\)](#)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

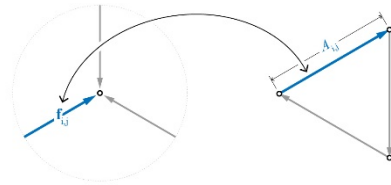
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

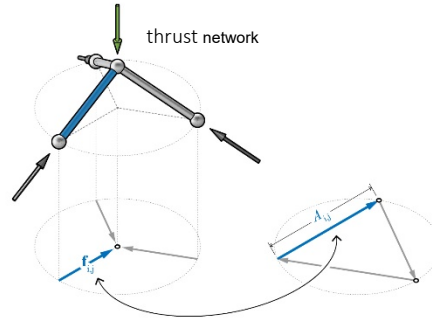
Combinatorial Equilibrium Modelling (CEM)



form diagram

force diagram

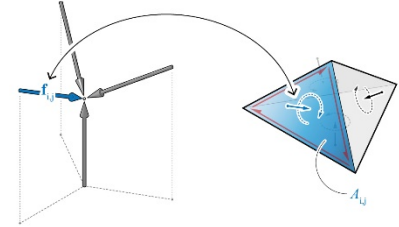
2D



form diagram

force diagram

2.5D



form diagram

force diagram

3D

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

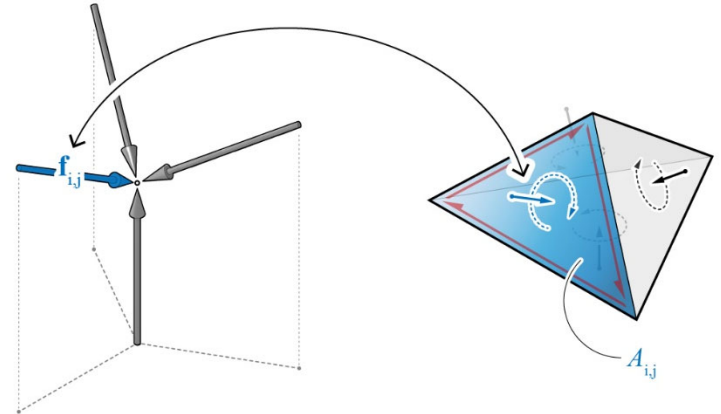
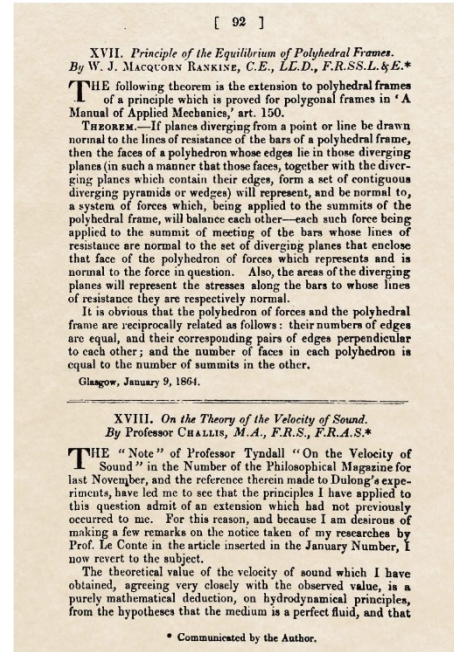
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

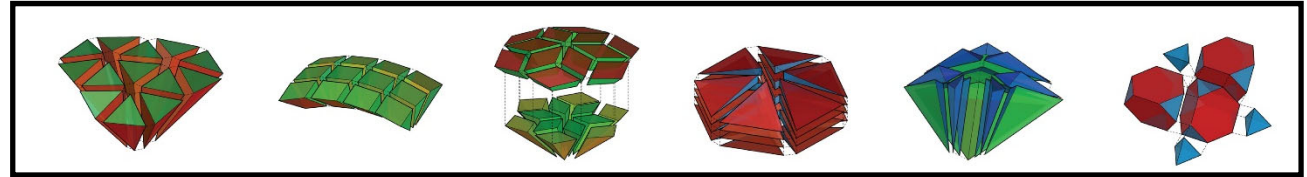
Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Polyhedral force diagram



Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)

Polyhedral form diagram



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

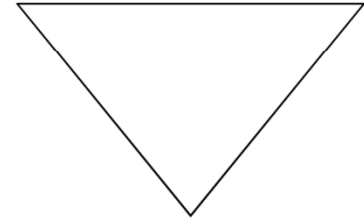
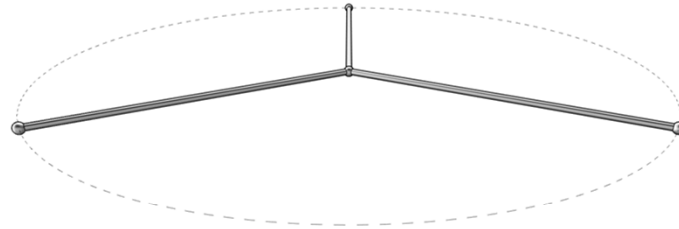
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

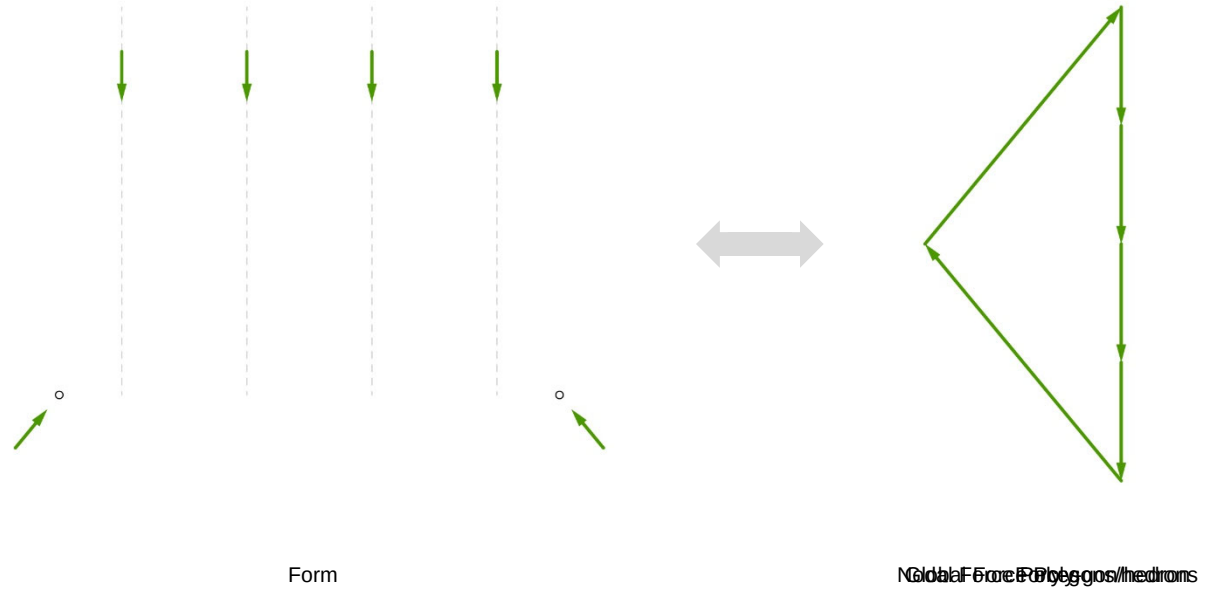
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

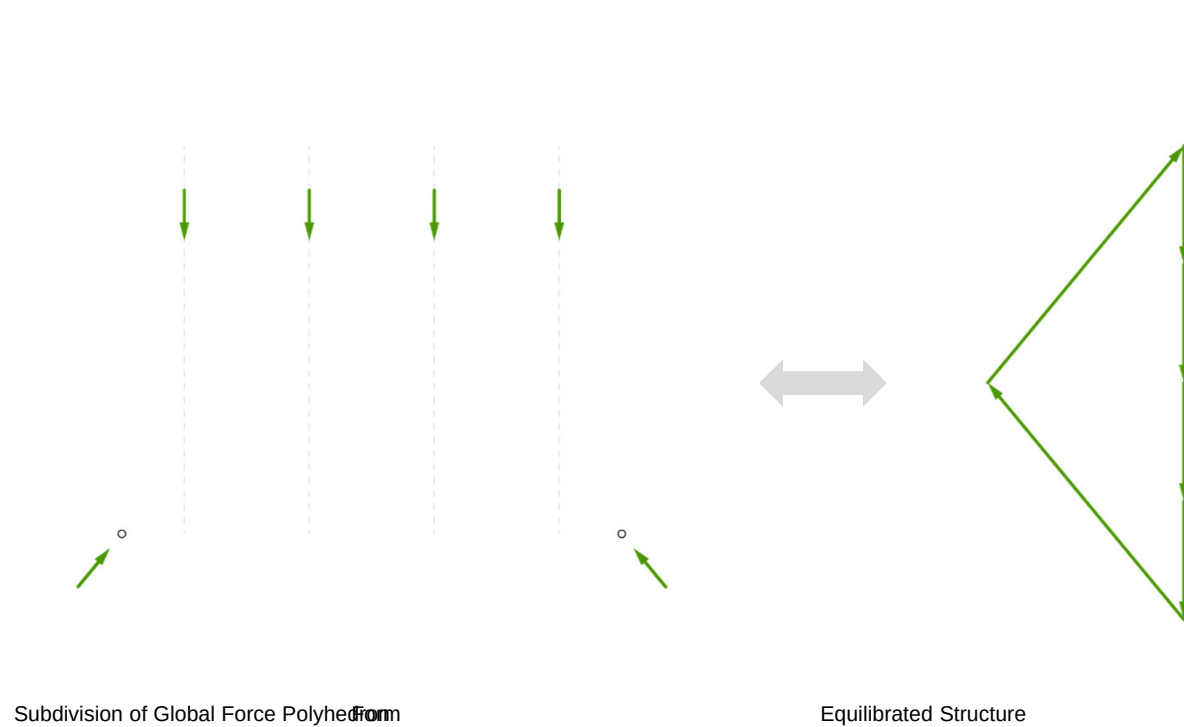
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

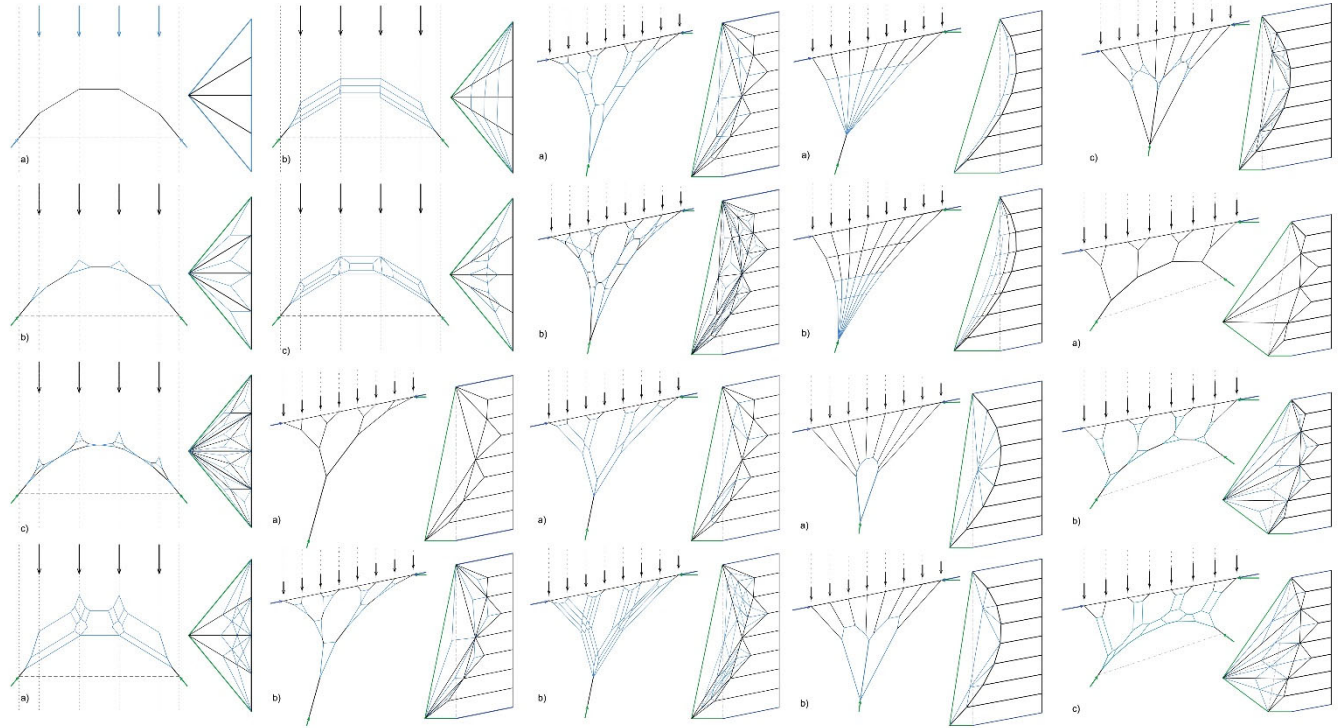
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



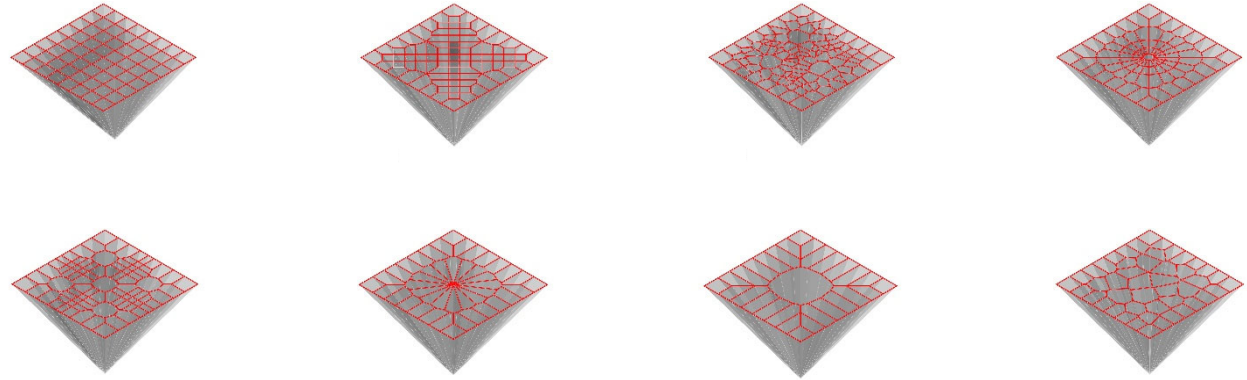
Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)



Form-finding

Interactive drawings

Constrained form-finding

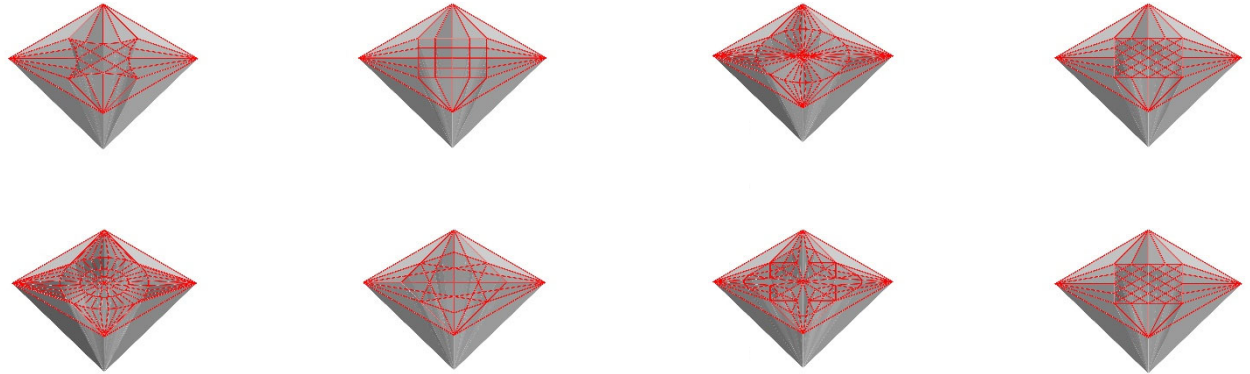
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

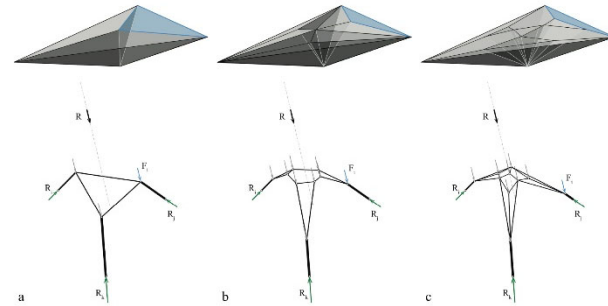
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

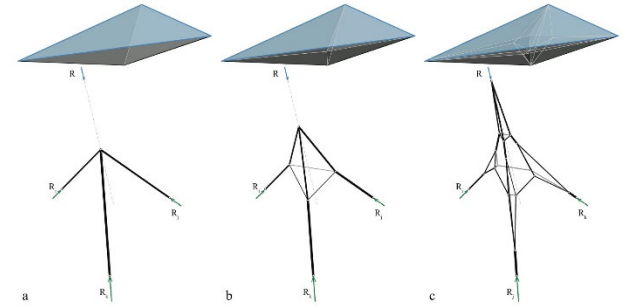
[Subdivision](#)

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Subdivision of a Face



Subdivision of Cells

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

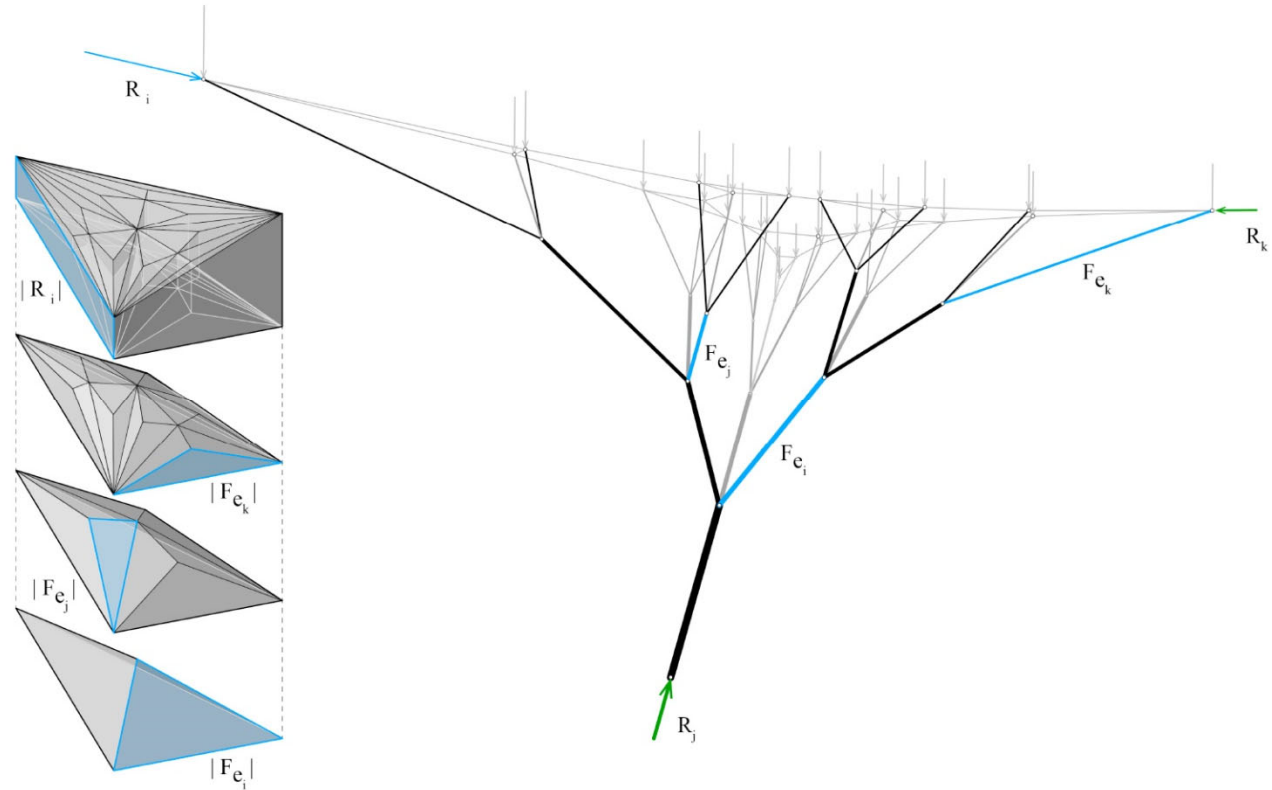
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

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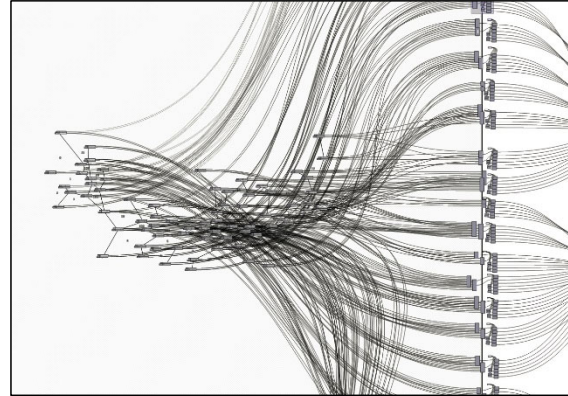
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

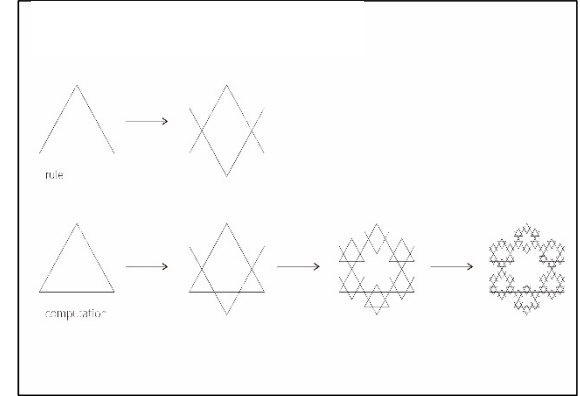
Combinatorial Equilibrium Modelling (CEM)



Parametric Design Space

NUMERICAL PARAMETERS are defined ...
to
... DESCRIBE & MODIFY geometries.

VS



Grammatical Design Space

ACTIONS & OPERATIONS (RULES) are defined ...
to
... GENERATE geometries.

Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

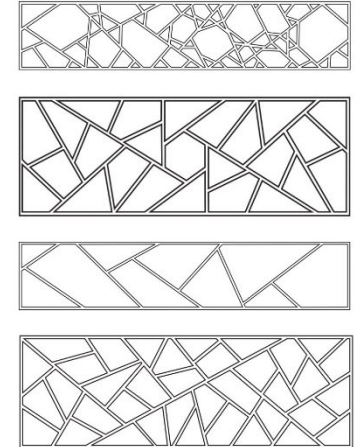
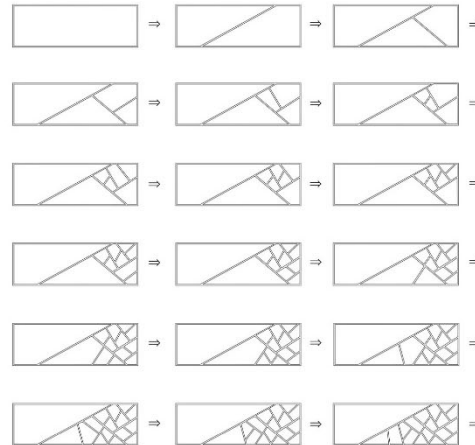
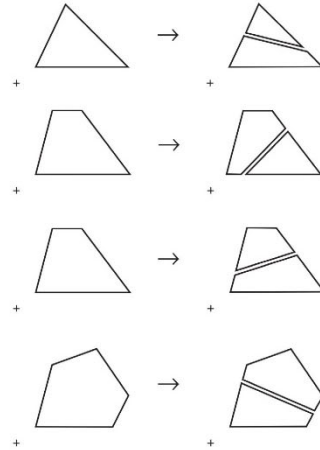
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

[Shape grammars](#)

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

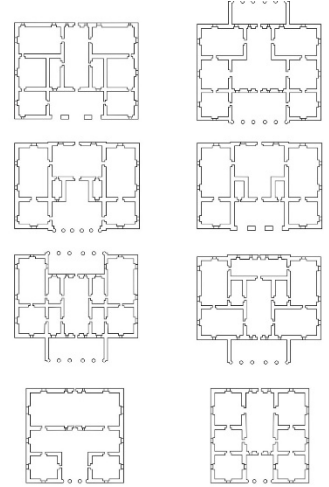
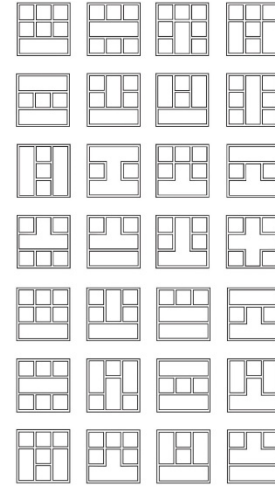
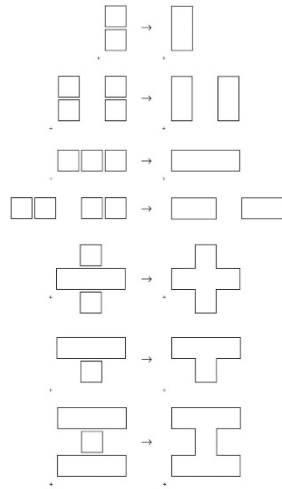
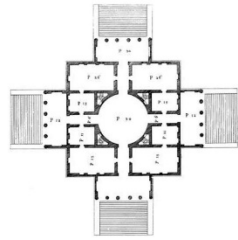
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Interactive drawings

Constrained form-finding

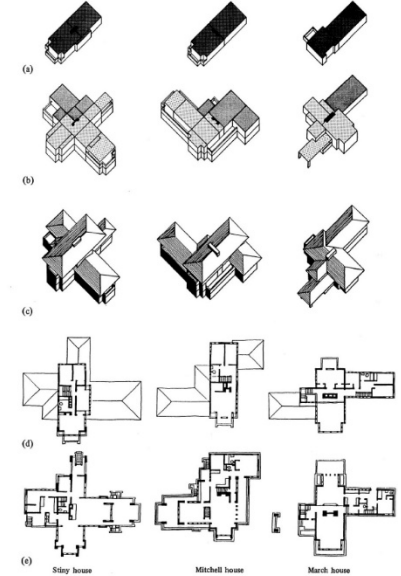
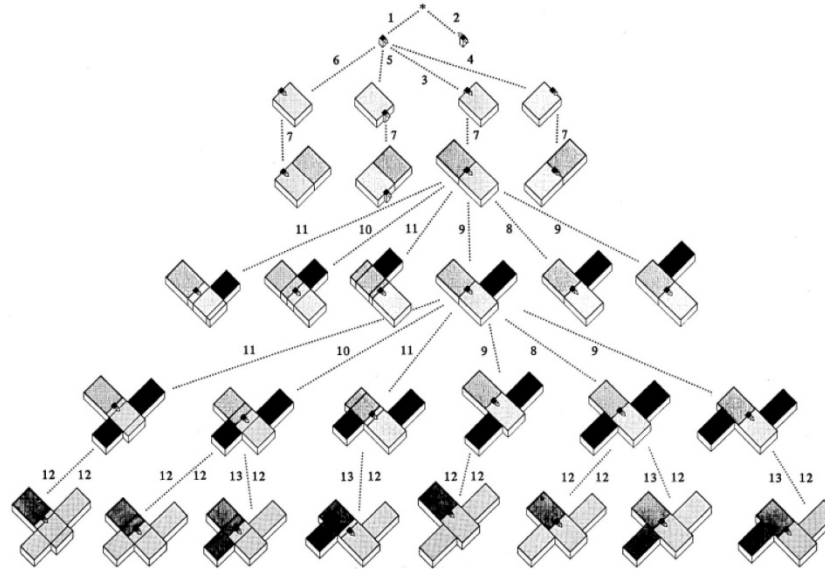
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

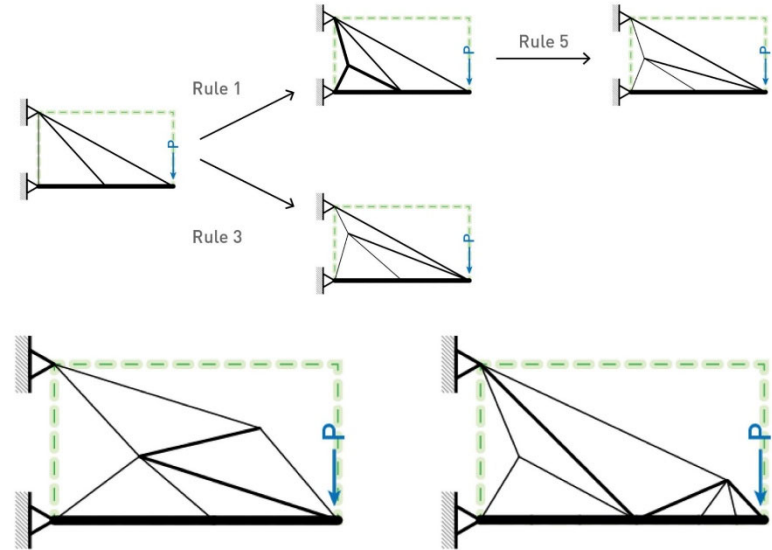
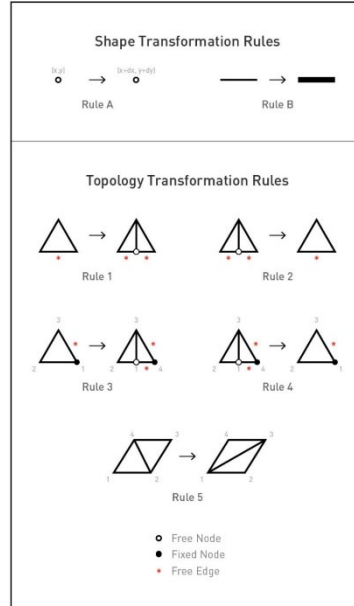
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Original Work by Shea & Cagan (1999) • Redrawn & f

Analysis

Interactive drawings

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Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

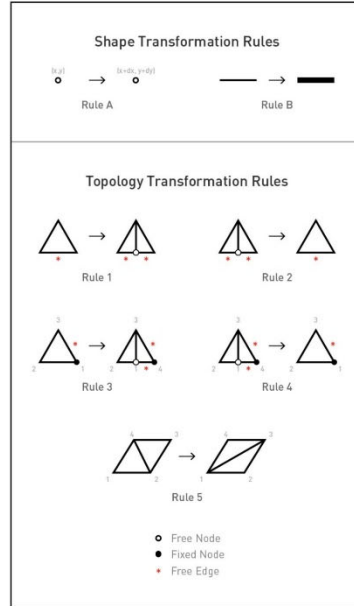
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

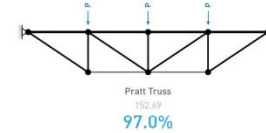
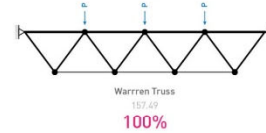
Combinatorial Equilibrium Modelling (CEM)



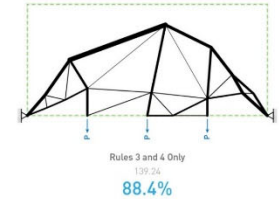
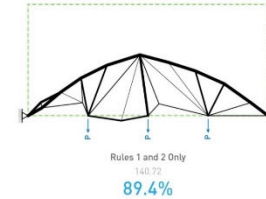
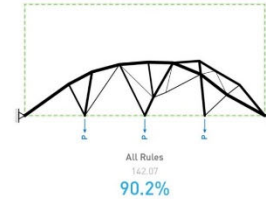
Design Problem



Conventional Solutions



Shape Grammar Solutions



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

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Interactive drawings

Constrained form-finding

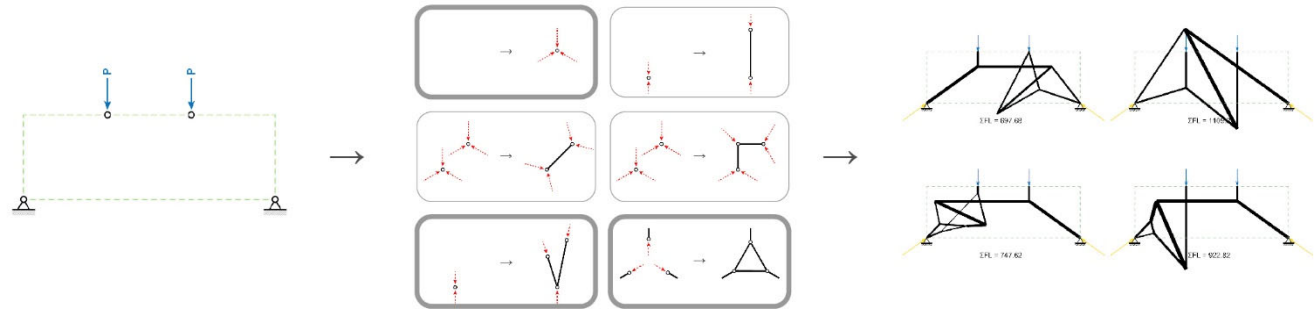
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

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Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

[Shape grammars](#)

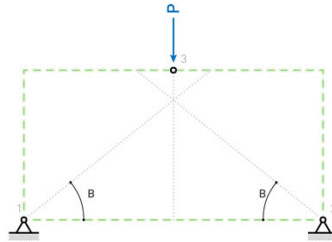
Combinatorial Equilibrium Modelling (CEM)

GLOBAL EQUILIBRIUM

—

TEMP FORCES

—



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

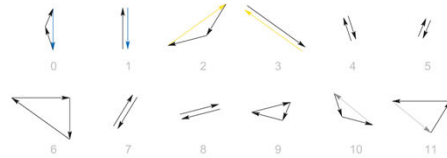
3D Graphic Statics (3GS)

Subdivision

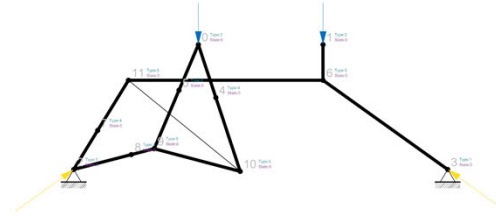
Shape grammars

Combinatorial Equilibrium Modelling (CEM)

Force Diagrams of Every Node



The Design



Evaluation Metric

ID : 0004E

$\Sigma FL = 673.61$

$\Sigma L = 126.38$

$\Sigma \# = 13$

Rule 2 25%
Length: 1.00
Angle: 150.00/15.00
Rule 3 25%
Length: 1.00
Rule 4 25%
Length: 1.00
Rule 5 25%
Length: 1.00

Rule Info

RULE HISTORY

1. "Rule 0: START" applied at Node 2
2. "Rule 3: Split Node" applied at Node 0
3. "Rule 5: Connect via Projection" applied at Node 4
4. "Rule 5: Connect via Projection" applied at Node 1
5. "Rule 3: Split Node" applied at Node 2
6. "Rule 5: Connect via Projection" applied at Node 8
7. "Rule 6: System Close" applied at Node 4
8. "Rule 6: System Close" applied at Node 10
9. "Rule 6: System Close" applied at Node 6

Rule History

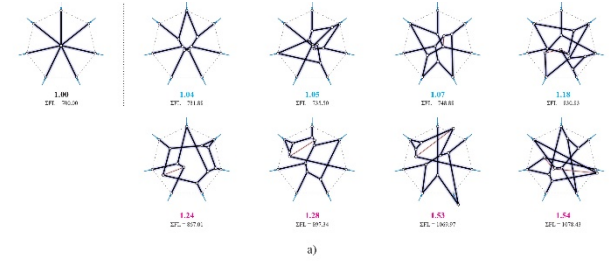
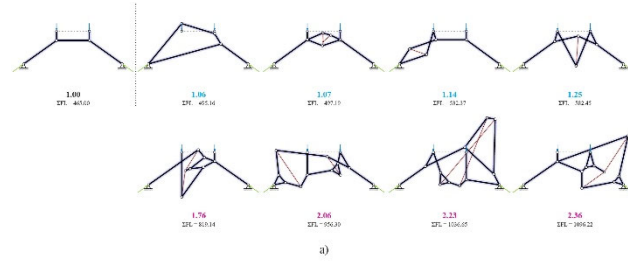
Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)



Form-finding

Interactive drawings

Constrained form-finding

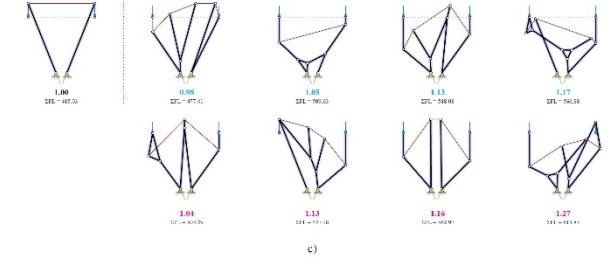
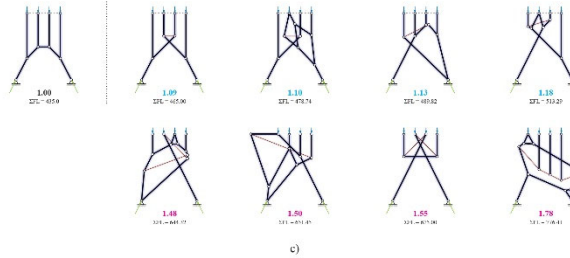
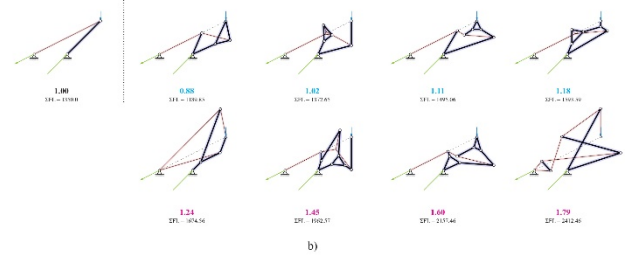
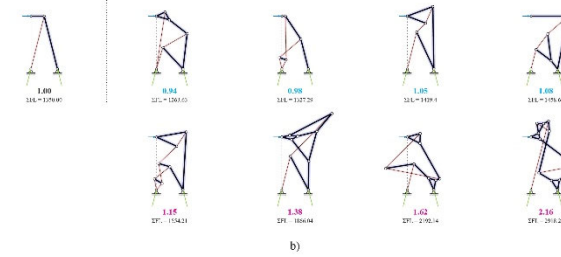
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



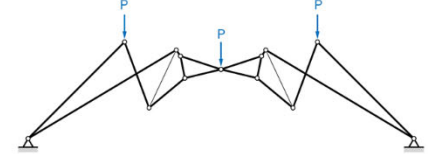
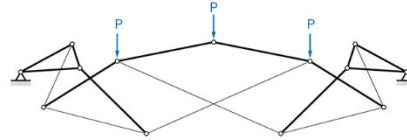
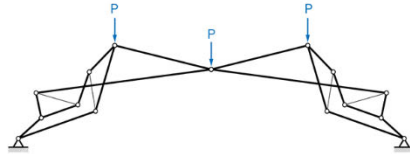
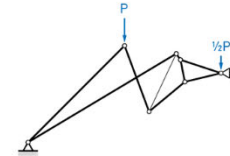
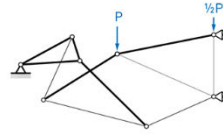
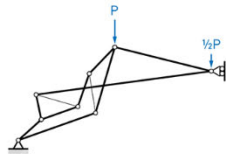
Analysis

Interactive drawings

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Form-finding

Interactive drawings

Constrained form-finding

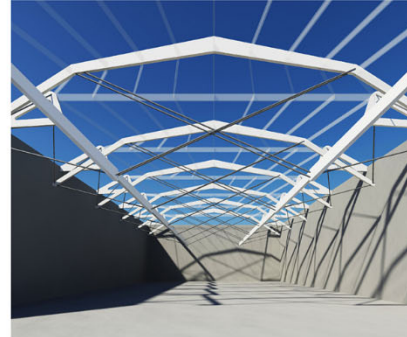
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

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Bi-directional AGS

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Form-finding

Interactive drawings

Constrained form-finding

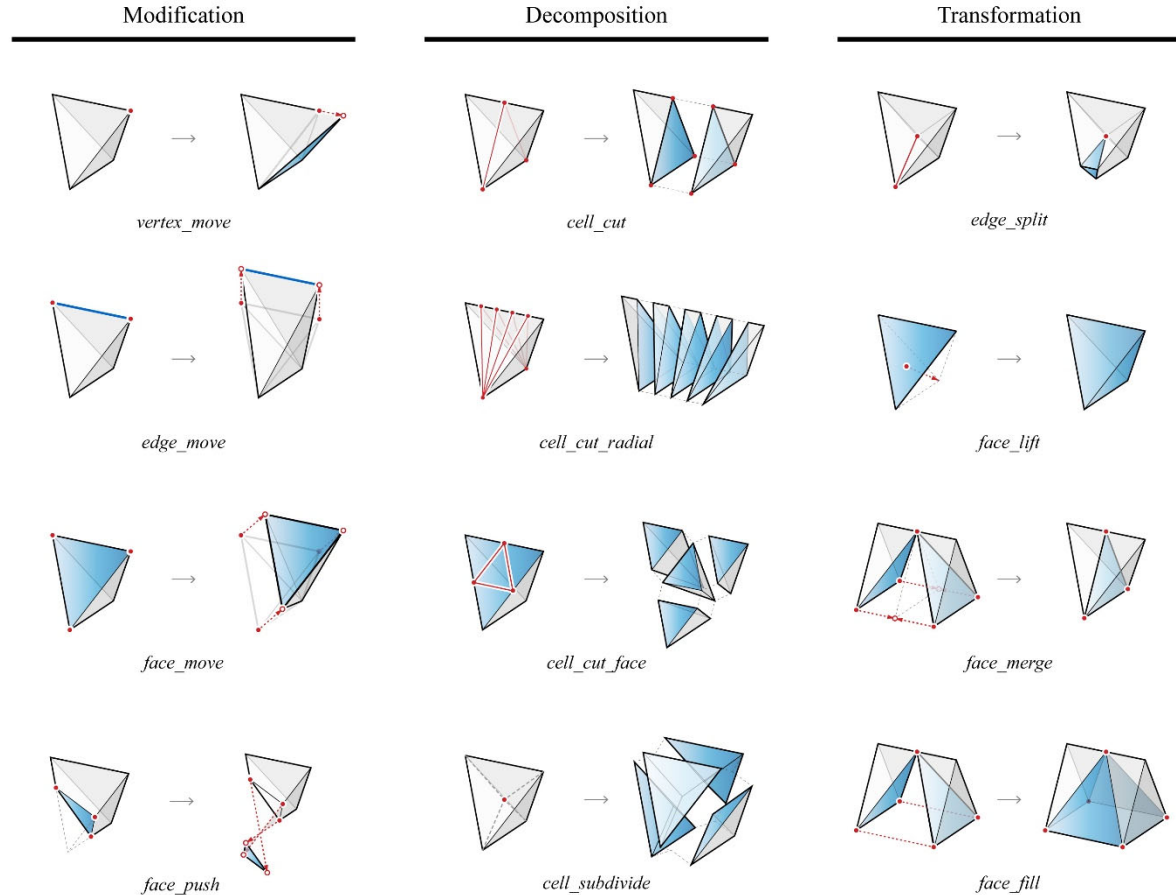
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

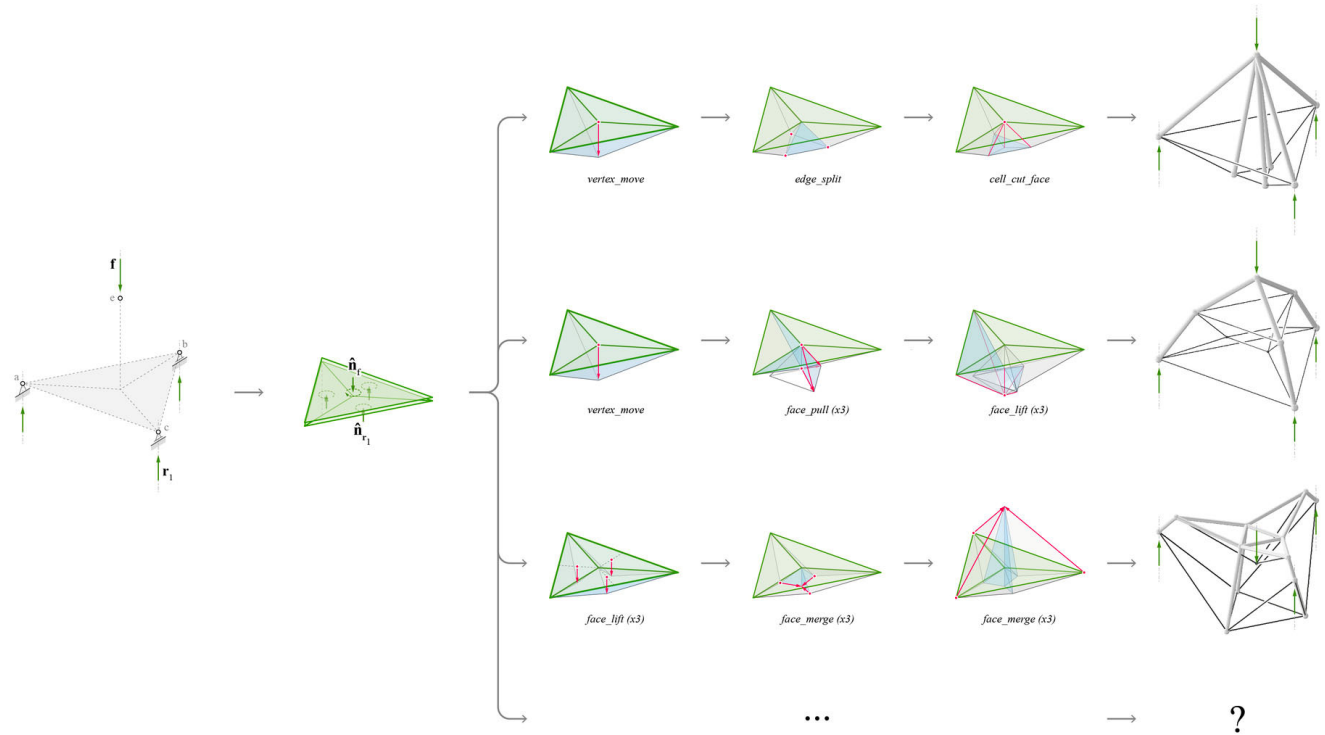
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

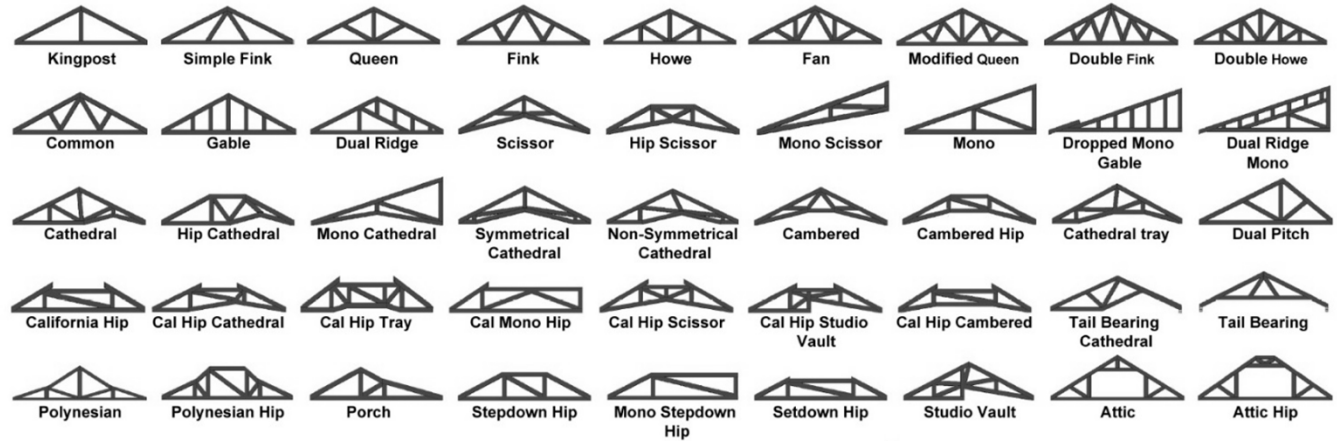
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



“In order to arrive at new forms and concepts,
we have to free ourselves from such pre-defined typologies.”

Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

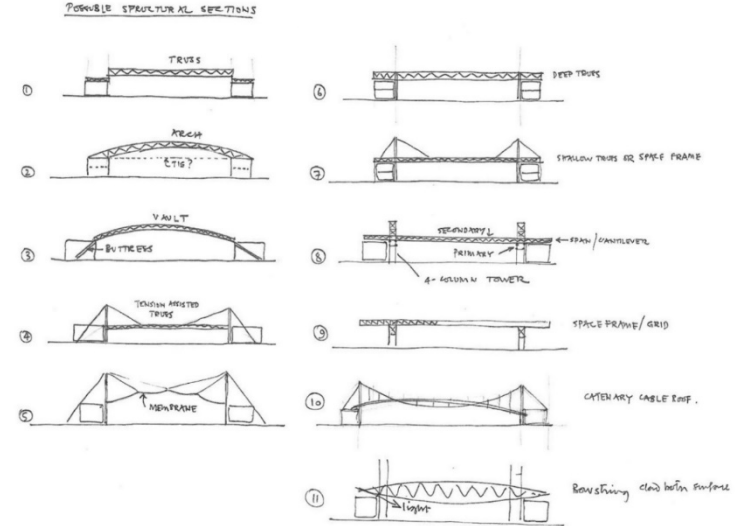
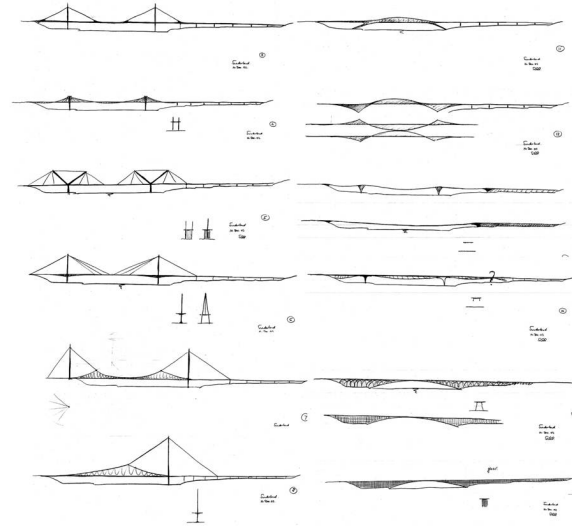
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

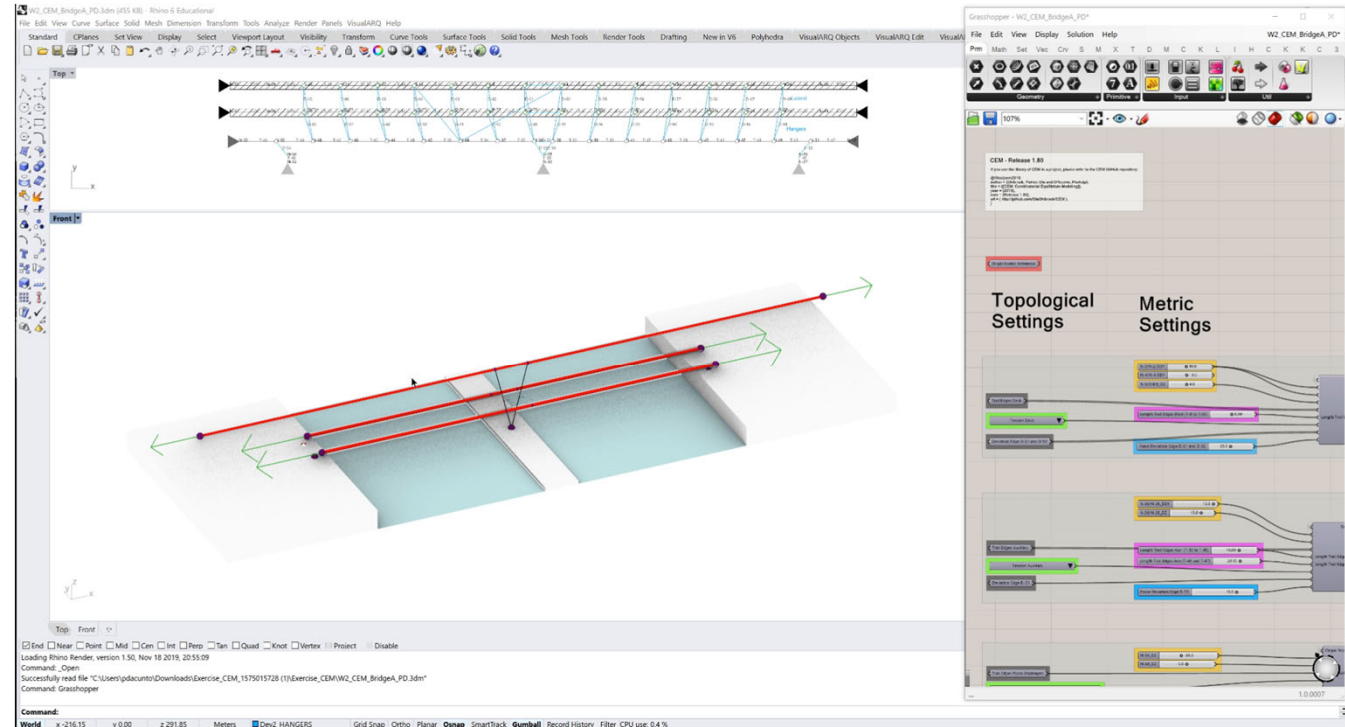
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

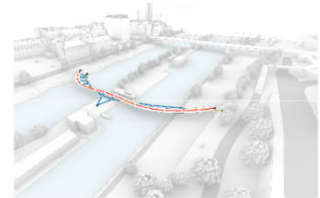
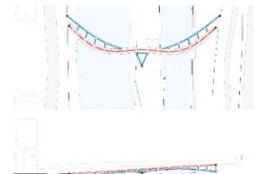
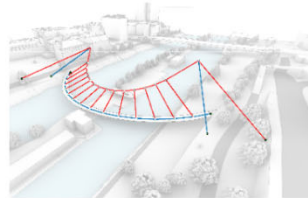
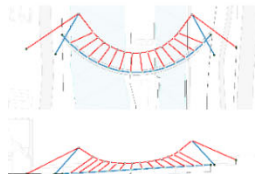
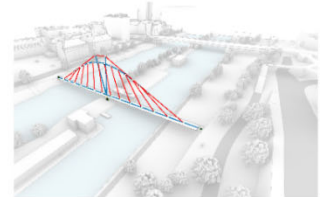
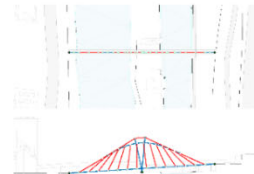
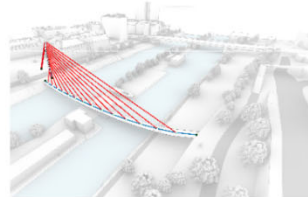
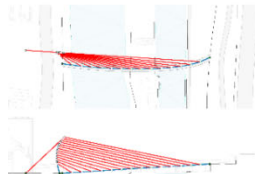
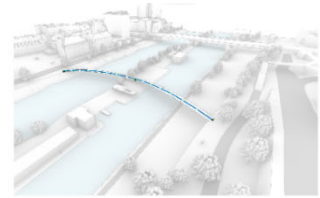
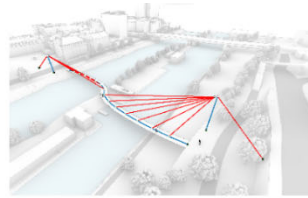
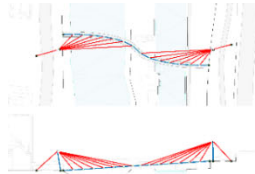
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

[Combinatorial Equilibrium Modelling \(CEM\)](#)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

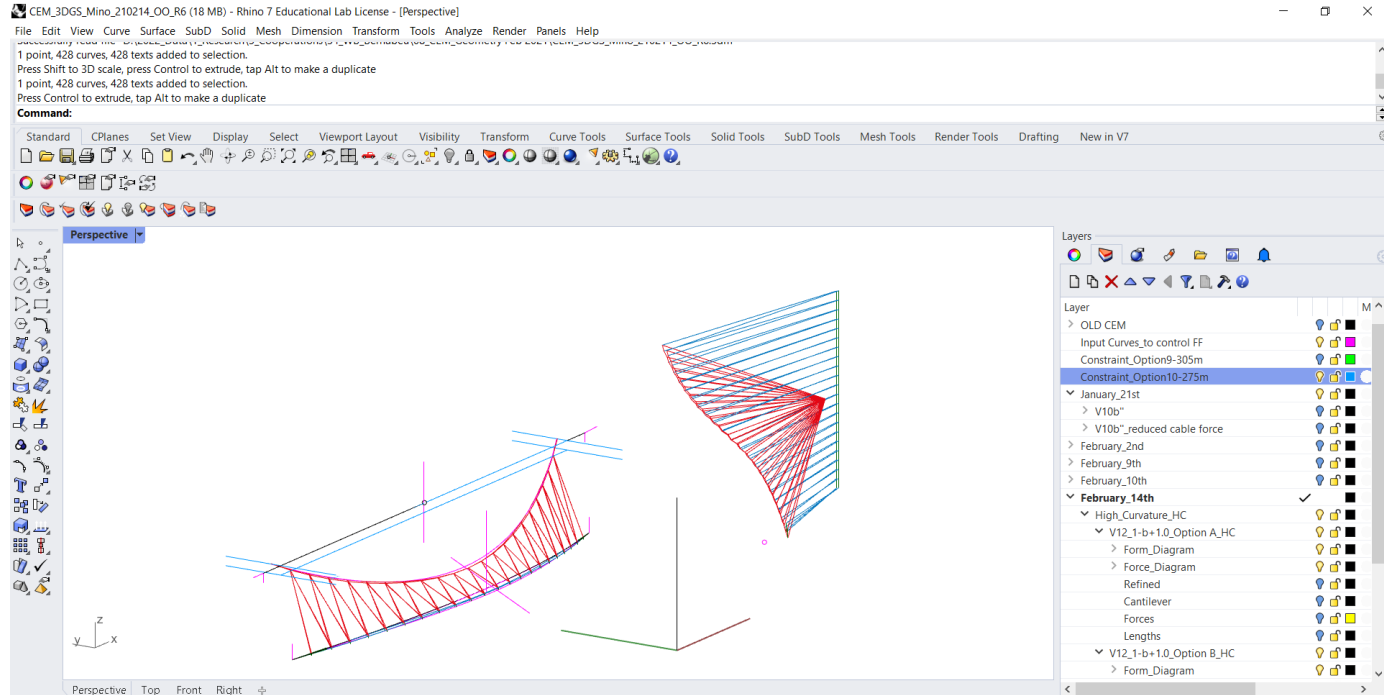
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

[Combinatorial Equilibrium Modelling \(CEM\)](#)



Analysis

Interactive drawings

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Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

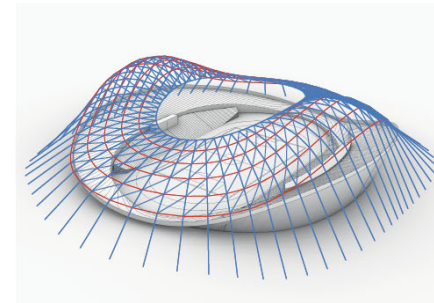
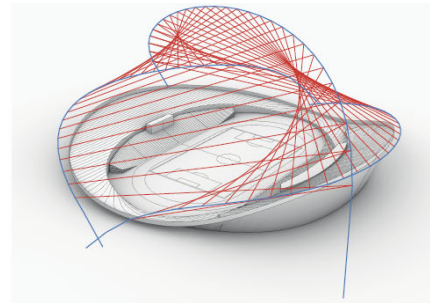
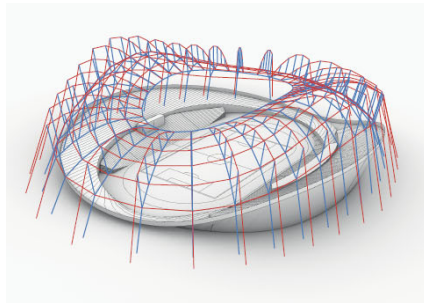
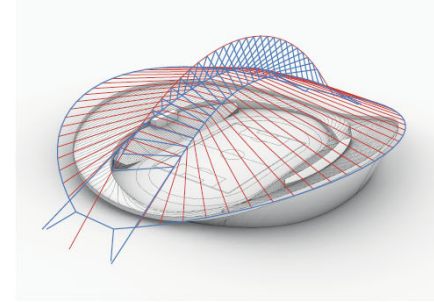
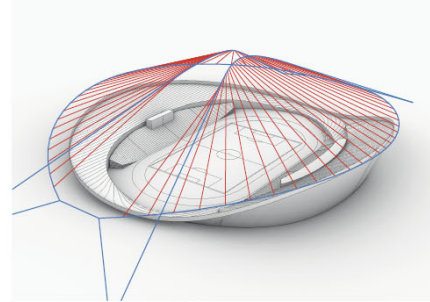
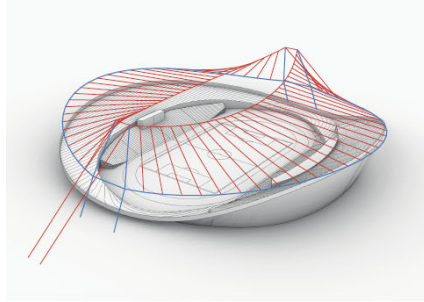
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

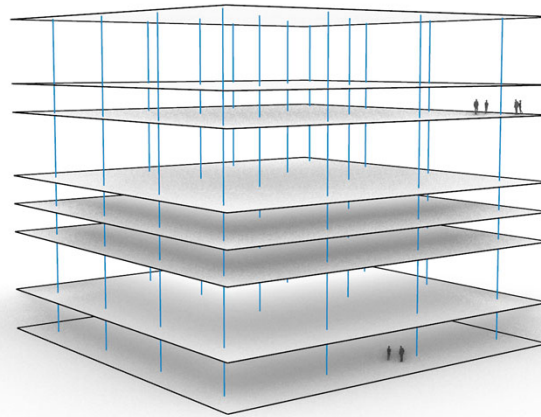
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

[Combinatorial Equilibrium Modelling \(CEM\)](#)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

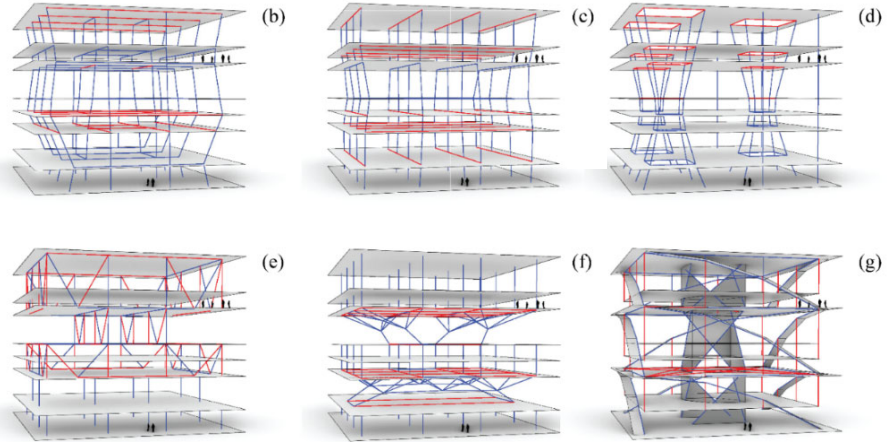
Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

[Combinatorial Equilibrium Modelling \(CEM\)](#)



Analysis

Interactive drawings

Algebraic graph(ic) statics (AGS)

Bi-directional AGS

Vector-based Graphic Statics (VGS)

Form-finding

Interactive drawings

Constrained form-finding

Thrust Network Analysis (TNA)

3D Graphic Statics (3GS)

Subdivision

Shape grammars

Combinatorial Equilibrium Modelling (CEM)

